

CYCLOTRON LINE FEATURES FROM NEAR-CRITICAL MAGNETIC FIELDS: THE EFFECT OF OPTICAL DEPTH AND PLASMA GEOMETRY

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ABSTRACT

A detailed analytical study of the relativistic profile for magnetic Compton scattering is performed and use is made of Monte Carlo methods to construct a numerical scheme for the production of relativistic theoretical cyclotron spectra from internally irradiated scattering atmospheres in the low-density/high-field regime. The cross section includes natural line widths, and the Monte Carlo implementation allows for spatial diffusion of photons in arbitrary geometries while accounting for relativistic angular redistribution. Detailed consideration of “Raman-Landau” scattering permits *photon spawning* from up to fourth harmonic photons. This scheme is utilized to make specific predictions on the effect of geometry and optical depth on the formation of cyclotron lines in hard incident continua. Even if the injection is isotropic, a strongly anisotropic radiation field ensues at relatively low optical depths. The scattering profiles at large angles and low fields are highly asymmetric and peak at the relativistic energy cutoff, not necessarily at the resonant energy. Higher harmonic features in general bear little resemblance to the fundamental even at high fields ($B \sim 10$ TG), and slabs yield the deepest features, while cylinders present the widest lines at small angles. Moreover, the fundamental line photons in slab scattering regions tend to redistribute to the line wings from all angles, producing fairly broad emission line wings at small angles. Higher optical depths at those angles hampers such redistribution in cylindrical geometries.

Subject headings: atomic processes — line: formation — magnetic fields — radiation mechanisms: nonthermal — scattering — stars: neutron

1. INTRODUCTION

Neutron stars have long been argued to possess magnetic fields of $\mathcal{O}$ (TG) on grounds of magnetic flux conservation during the gravitational collapse of supernovae cores (Woltjer 1964; Hoyle, Narkilar, & Wheeler 1964; Pacini 1967). In fact, fields nearly as high as the “critical”³ QED field scale, $B_{\text{crit}} = m^2/e(c^3/\hbar) \simeq 44$ TG, in some pulsars are evidenced by high spin-down rates attributed to magnetic dipole radiation (Gold 1968; Ostriker & Gunn 1969). Theoretically, still higher, “supercritical” magnetic fields are predicted to occur during vigorous entropy driven convection episodes of very young neutron stars (Thompson & Duncan 1993), and the recent measurement of a very fast spin-down rate⁴ for soft gamma repeater SGR 1806-20 (Kouveliotou et al. 1998) supports the existence of at least *one* such magnetar. As magnetars may also manifest themselves as anomalous X-ray pulsars (Thompson & Duncan 1996), the presence of supercritical magnetic fields may not be as rare as was once thought in spite of the lack of spectral evidence.

More direct observational evidence of the magnetic field strength in neutron star sources is provided by X-ray cyclotron “lines.” These quasi-harmonic spectral features are generated by the scattering of resonant photons with the electrons embedded in the magnetospheres of such objects. The interaction is resonant because the Dirac equation in the presence of a locally uniform magnetic field, $\mathbf{B}$, yields discrete perpendicular momentum eigenvalues: $p_{\perp}/m_e = n(B/B_{\text{crit}})$ (e.g., Melrose & Parle 1983), although relativistic

effects introduce a slight anharmonicity in the rest frame resonant photon energies:

$$\omega_n/m_e = \{[1 + 2n(B/B_{\text{crit}}) \sin^2 \theta]^{1/2} - 1\}/\sin^2 \theta. \quad (1)$$

As these energies depend nontrivially on the photon’s propagation angle with respect to the field, $\theta = \cos^{-1}(\hat{\mathbf{k}} \cdot \hat{\mathbf{B}})$, the emergent spectral features are influenced heavily by the spatial distribution of electron targets (i.e., plasma “geometry”) and the scattering region’s orientation with respect to the magnetic field.

Cyclotron lines have been detected in a number of X-ray pulsar spectra (Nagase 1989; Mihara 1995) and possibly also in the spectra of several gamma-ray bursts (GRBs; Mazets et al. 1981, 1983; Murakami et al. 1988; Yoshida et al. 1991). While there is a broad consensus on the origin of the former objects, the origin of GRBs is still elusive. Line energies in X-ray pulsars typically fall in the range from 4 to 40 keV, indicating scattering of electrons in magnetic fields of up to 3.5 TG. The case of transient X-ray pulsar A0535+26, with an 11 TG field ($\sim \frac{1}{4}B_{\text{crit}}$), represents the uppermost limit spectrally determined thus far (Grove et al. 1995; Araya & Harding 1996a, 1996b; Maisack et al. 1997).

The interpretation of very high energy X-ray spectra is a highly complex task, particularly on the matter of judging the veracity of cyclotron lines. On one hand, observational photon spectra at energies in excess of ~ 10 keV are the result of a procedure that assumes an empirical spectrum and convolves it with the instrument’s response function (for instance, through a forward folding deconvolution procedure), producing a reduced χ^2 as a measure of self-consistency. The deduced spectra do not yield “rigid” data points for the theoretician to interpret or model. On the other hand, formal numerical models of theoretical spectra from highly magnetized plasmas, because of their complexity and computational expense, do not readily lend themselves for the exhausting trial and error task that the experimental procedures require. Furthermore, the intrinsic

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³ This misnomer is derived from the fact that an equivalent strength electric field leads to *Klein’s* catastrophe: a uniform magnetic field *effectively* forces the Dirac sea into a magnetized virtual state instead.

⁴ $\dot{P} \simeq 2.3 \times 10^{-3}$ s yr⁻¹ and $n_{\text{break}} = 3$ yield an inferred field strength $B \sim 8 \times 10^{14}$ G.

spectra of X-ray pulsars (the most likely candidates for detection) tend to turn over at energies larger than ~ 20 keV, thus making line detections above this turnover difficult to obtain.

The production of theoretical cyclotron line features in an entirely self-consistent manner is a formidable task requiring the resolution of a coupled radiation-magnetohydrodynamic system (see, for instance, Arons, Klein, & Lea 1987). Even simplified approaches pose extremely difficult problems, and the solutions are typically attempted in piecemeal form. In idealized geometries, the radiative transfer equation is cast into a two-point boundary value problem with numerically viable solutions. Alternatively, a less traditional approach to produce the spectra is through a Monte Carlo program (MCP), as in the calculation we describe in detail here. In either case, the physical state of the plasma in the line-forming region and its macroscopic shape or “geometry” are prescribed.

Theoretical cyclotron spectra have been investigated in this simplified manner by very many groups, and we will make no attempt to produce a comprehensive bibliographic list. For cyclotron features produced in hard incident continua (as in this investigation), the work led by the groups of Isenberg, Lamb, & Wang (1997; Lamb et al. 1989; Wang et al. 1989; Wang, Wasserman, & Salpeter 1988; Wasserman & Salpeter 1980; Ventura, Nagel, & Mészáros 1979) and of Alexander & Mészáros (1989; Alexander, Mészáros, & Bussard 1989; Bussard, Alexander, & Mészáros 1986; Mészáros & Nagel 1985a, 1985b; Nagel 1981a, 1981b) depict the state of the art calculations and are most closely related to the models produced here.

Isenberg et al. (1997) offer a thorough review of the general problem of cyclotron line formation with regard to the line photon transfer. Radiative transfer through slab geometries is implemented through a Monte Carlo scheme, and the transfer of both polarized and unpolarized photons is considered. However, use of a nonrelativistic cross section with relativistic kinematics, although extremely useful in complex calculations, limits the scope of their work to systems where $|B| \ll B_{\text{crit}}$.

Alexander & Mészáros (1989), on the other hand, used relativistic cross sections and a discrete, ordinate Feautrier method to iteratively solve a photon polarization mode-dependent radiative transfer equation. This scheme is numerically advantageous, as the radiative transfer equation is found to converge rapidly in a few iterations. However, use of the two-stream approximation allows for little angular diffusion, $\theta \rightleftharpoons \pi - \theta$, and this results in angle-averaged spectra, as the authors remarked.

A more complex Feautrier-type scheme that accounts for angular redistribution onto four angle bins ($0 \leq \theta \leq \pi/2$) was developed by Mészáros & Nagel (1985a), and this scheme was updated with a relativistic cross section to model X-ray pulsars’ magnetospheres by Bulik et al. (1992), including the effect of field curvature in homogeneous media while approximating the coupling of second harmonic line photons to the fundamental with a line profile at that energy.

Furthermore, Bulik et al. (1995) produced detailed cyclotron line fits for 4U 1538–53 and Vela X-1 while accounting for general relativistic effects and computing a (parallel) temperature gradient in radiative equilibrium from ram energy deposition at the bottom of the accretion column. These represent major improvements to the Feautrier treat-

ment, but the applicable field strengths are severely limited by a use of the nonrelativistic cross section⁵ corrected to first order for relativistic kinematics. In addition, book-keeping of just eight angle bins: $\epsilon \in [0.0, \pi]$, makes treating the fine energy variation in the multiple resonances difficult.

The results presented in § 6 ensue from a hard incident continuum throughout the entire energy range (see eq. [61]). Although GRB sources generally possess much harder spectra than X-ray pulsars, some X-ray pulsars also exhibit a fairly hard X-ray continuum at the characteristic cyclotron energy (i.e., X0115+634, X0331+53, 4U 1538–52, and even A0535+26 at 110 keV). Theoretical modeling of cyclotron lines in such hard continua is difficult. The formation of the fundamental cyclotron line feature is strongly coupled to the radiation field at higher harmonic energies through photon spawning (§ 5). Our aim in producing the cyclotron line spectra in hard continua is twofold. First, such X-ray continua shone on magnetized plasmas allows for an easier interpretation of the effects of photon spawning, geometry, and relativistic angular redistribution of resonant photons on the formation of the fundamental cyclotron line feature. The theoretical spectra thus generated greatly aid in the interpretation of spectra formed in softer continua such as those belonging to X-ray pulsars (the effect of photon spawning will be weaker in those sources, however). Second, as this study was originally motivated by the double cyclotron lines reported for GB 880205 by the *GINGA* satellite, we make a qualitative comparison with the models of Wang et al. (1989).

In the next section a synopsis of the model is presented, and reference is made to the underlying physical assumptions that delimit the regime of validity of our formulation. The assumed astrophysical setting for cyclotron line formation is briefly reviewed in § 3 and the explicit form of the cross section given in § 4. The implementation of a Monte Carlo scheme for the scattering process and photon transfer, along with a detailed study of the relativistic profile, are discussed in § 5. This formulation is employed to produce theoretical cyclotron spectra from hard incident continua in § 6, paying particular attention to the optical depth progression of line features and to the effect of geometry. Conclusions are presented in § 7.

2. PHYSICAL ASSUMPTIONS AND SYNOPSIS OF THE MODEL

The central issue in our investigation is the construction of a numerical model tailored to tackle the radiative transfer of unpolarized continuum photons in a strongly magnetized, low-density plasma. Such a program also permits a fairly accurate treatment of an astrophysical setting with a *near-critical* magnetic field (but still with $|B| < B_{\text{crit}}$). In fact, the low-density/high-field regime represents a chief simplification to our treatment of the problem. Motivating this restriction is the assurance that vacuum polarization determines the photon states (§ 3.3.1), collisional interactions may be ignored (§ 3.2), and the transfer of unpolarized photons will give a reasonable description of the emergent radiation field (see below).

The relativistic, magnetic Compton-scattering cross section includes natural line widths through the use of dressed electron propagators evaluated in the presence of a uniform magnetic field. The electron states that permit such

⁵ Resonant only at the fundamental cyclotron energy.

construction are simultaneous eigenstates of the Sokolov-Ternov relativistic spin operator, § 3.3.2, and the photon polarization states are the magnetized vacuum normal modes uncorrected for dispersion, § 3.3.1.

Detailed consideration of “Landau-Raman” scattering permits photon spawning from up to fourth harmonic photons, § 5.5, and the scattering process, in general, samples the electron momenta from a relativistic thermal energy distribution function, § 5.3. A thermal characterization implies the plasma is understood to be an electron-baryon mix. Moreover, all electrons are initially in the ground Landau level, § 3.2, so that the temperature corresponds to the “parallel” component of the electron momenta.

Furthermore, a Monte Carlo implementation for photon injection, propagation, and scattering permits the spatial diffusion of photons from arbitrary photon source functions and in arbitrary plasma geometries. A relativistic photon angle redistribution function, accurate for core scattering events that leave the electron back in the ground level and approximate otherwise, is also used to allow for angular diffusion, § 5.6.

While an exact treatment of the electron-photon interactions would be ideal, an attempt is made to strike a balance between accuracy and the reasonable use of computational resources. The “on-resonance” approximation used to sample angular redistribution and choose the target electrons’ initial momenta in the majority of scattering events is strictly valid as long as the resonant behavior of the cross section dominates (§§ 4.3 and 5.3.1). Accuracy in the line core is warranted but not in the wings, where the scattering cross section provides a more accurate description.

A generalization of the code to account for photon polarization in the transfer problem is a highly desirable feature, but this is not a central issue in this investigation. In the strong field limit, use of the normal photon modes yields mode-dependent cross sections that are not substantially different from one another near the resonance and, mode-switching branching ratios strongly couple the two modes. On the other hand, if plasma polarization modes are used the cross section for the ordinary mode is not resonant unless one includes spin-flip transitions (Bonazzola, Heyvarets, & Puget 1979). Thus, Wang et al. (1988) found that both modes, $\parallel$ and $\perp$, have comparable mean free paths and contribute about equally to the spectral angular output distributions in slab geometries. The assumption that the deviations of the unpolarized from the polarized treatment are minimal renders the numerical code computationally manageable.

The spectra are formed by unbiased injection of individual photons from N_E identical energy bands evenly distributed throughout the spectral range of interest. This is followed by reprocessing of each photon through scattering and subsequent collection of the escaping radiation in a similar energy grid. To avoid poor statistics at high energies, photons are injected from a flat spectrum and assigned weights that depend on their energy and the spectral index. Two geometries for the scattering region are considered (§ 3.1): plane parallel slab (with the magnetic field parallel to the slab’s normal vector) and cylindrical (with the field parallel to the cylinder axis). Because of axial symmetry, the $\hat{\mathbf{z}}$ direction is oriented along the magnetic field $\mathbf{B}$. A continuum photon spectrum is incident from a source at the

slab midplane or the cylinder axis. Internally, the source code keeps track of $N_\mu = 20$ equally spaced values of $\mu \equiv \hat{\mathbf{k}} \cdot \mathbf{B} \in [-1, 1]$, the cosine of the viewing angle to the field, and $N_E = 80$ energy bins with $k^0 \in [\omega_{\min}, \omega_{\max}]$. The injected and escaping photons are accumulated into four ranges of $\mu \in [0, 1]$, and angle-dependent spectra are formed by considering the number of emerging photons into each of the N_E energy bins. The Thomson (continuum) optical depth of the plasma in the line forming region, the electron temperature, and the spectral index of injected photons are prescribed.

3. THE ASTROPHYSICAL SETTING

The cyclotron line spectra are produced (§ 5) in *irradiated* magnetized plasmas without internal photon sources and with *prescribed* thermodynamic conditions. The astrophysical setting of X-ray pulsars is discussed next to motivate the latter prescription.

3.1. Plasma Geometry

X-ray pulsars are found in binary systems containing a neutron star. In high-mass X-ray binaries, a tenuous accretion wind occurs from the massive companion onto the neutron star surface. Low-mass systems have higher accretion rates and disks surrounding the neutron star (see van den Heuvel & Rappaport 1987 for a review). The mechanism for flow stopping at the surface of the star is not certain in either case. However, if shocks form radiatively in high-luminosity sources (Basko & Sunyaev 1976; Davidson 1973; Wang & Frank 1982; Arons, Klein, & Lea 1985, 1987) or through a collisionless plasma instability in the flow of low-luminosity sources (Langer & Rappaport 1982), the geometries of the X-ray emitting region are most likely surmised as cylindrical columns. This situation sharply contrasts the case of flow stopping through Coulomb collisions with the ambient electrons, where plane parallel slabs with scale heights $h \sim 200$ cm are expected (Mészáros et al. 1983; Harding et al. 1984).

These possible pictures of the flow provide two limiting “orthogonal” geometric scenarios: slabs and cylinders. As we shall see, our models of *cyclotron line features* may, in fact, provide a sensible diagnostic to determine the geometry of the region where the lines form. Consequently, these models could also help answer some important questions regarding the microphysics involved in the stopping and cooling of the flow.

3.2. Plasma Thermodynamic State

The general problem of line formation in layered atmospheres (i.e., stellar surfaces) has been debated extensively in the literature and is comprehensively reviewed by Mihalas (1978; see also Mihalas & Mihalas 1984). The thermodynamic state of the material in the outer layers of the star is largely determined by radiative cooling and heating rates that, at low particle densities, depend on the global properties of the radiation field. Photon transport at optical depths of about unity connects spatial regions that are otherwise physically disconnected and thermodynamically inhomogeneous. The scattering terms in the radiative transfer equation exacerbate the nonlocality of the problem, as they represent noncoherent diffusion of radiation through the atmosphere. Thermal characterization is thus difficult to justify, but the resonant behavior of the magnetic Compton-scattering cross section ensures strong photon-

electron coupling (at least at resonant energies) and marginally permits such energy distribution prescription for electrons.

3.2.1. Excited Landau Populations

The Landau-level populations should be determined from the excitation and de-excitation rate equations, including collisions and radiative interactions. However, this detailed a calculation is not necessary if one considers plasmas that are tenuous and cool relative to the cyclotron energy.

The radiative cyclotron de-excitation rate is given by (Latal 1986):

$$\nu_r = \alpha m_e [c^2/\hbar] (B/B_{\text{crit}})^\beta \sim 2 \times 10^{15} B_{12}^2 \text{ s}^{-1}, \quad (2)$$

with $\beta \in [2, 0.5]$ for the subcritical and supercritical field limits, respectively. Comparing this with the expression from the collisional rate (Bonazzola et al. 1979),

$$\nu_c = 5 \times 10^8 (n_e/10^{21} \text{ cm}^{-3}) B_{12}^{-3/2} \text{ s}^{-1},$$

and considering that the continuum (Thomson) optical depths, $\tau_c \equiv \tau \times \sigma_{\text{Th}}/\sigma_{\text{scat}} \leq 3 \times 10^{-3}$, used here correspond to relatively low column densities, $N_l \leq 4.5 \times 10^{21} \text{ cm}^{-2}$, the neglect of collisions is readily justified at low (continuum) optical depths.

3.2.2. Temperatures

Standard estimates for T_e from the Doppler line width, $\Delta\omega_{\text{dop}} = \omega_{\text{cyc}}(2kT_e/mc^2)^{1/2} \cos \theta$, are inadequate for two reasons. First, relativistic kinematics produces asymmetric broadening even at $\theta = 90^\circ$. Second, fundamental harmonic features have complex structure resulting from angle mixing and extra injection from spawned photons. In practice, this means that the line widths of fundamental harmonic features do not lend themselves to a straightforward interpretation of the plasma temperature.

Detailed analysis of thermal balance in the line-forming region has been addressed by Lamb, Wang, & Wasserman (1990), assuming nonrelativistic magnetic resonant and nonresonant Compton cross sections. Following the single scattering formalism, assuming line dominated cooling and heating, and isotropic injection, their estimate for the electron temperature yields

$$T_e = \omega_{\text{cyc}}/(2 + \alpha_{\text{eff}}), \quad \tau \approx 1, \quad (3)$$

where $\omega_{\text{cyc}} = m_e B/B_{\text{crit}}$ is the nominal cyclotron frequency and with α_{eff} being (minus) the effective power-law index at the line. For $\tau \gg 1$, these authors produce a numerical estimate from Monte Carlo runs:

$$T_e \sim \frac{1}{4} \omega_{\text{cyc}} \quad \text{for} \quad \alpha_{\text{eff}} \sim 1, \quad \tau \gg 1. \quad (4)$$

Although we observed that the relativistic and nonrelativistic resonant cross sections for $B = 0.24 B_{\text{crit}}$ differ by as much as 50% one line width from resonance, the above estimates are used as guides yielding consistent results for hard incident continua (§ 6). On the other hand, the latter estimate of T_e is not justified for most X-ray pulsar models index because of softer continua.

3.3. The Strong Field Limit

3.3.1. Normal Photon States

An estimate of the plasma's influence on the photon polarization modes can be made by finding the frequency threshold above which the magnetic contribution to the

dielectric tensor of the medium dominates the plasma component. The collective influence of on-shell electrons is readily estimated by noting that away from the cyclotron frequency, the unit deviations of the dielectric tensor in the usual cold plasma approximation (and also for a relativistic electron plasma; Pavlov, Shibano, & Yakovlev 1980) scale quadratically with the “normalized” plasma frequency, $v \equiv (\omega_p/\omega)^2$, and that the deviations from unity due to vacuum polarization scale with $\varepsilon = \alpha/(45\pi)(B/B_{\text{crit}})^2$.

Thus for a plasma having $n_e \approx 10^{21} \text{ cm}^{-3}$, the lower threshold on photon frequency over which vacuum polarization rapidly begins to dominate the dielectric tensor is given by

$$\omega > \sqrt{\frac{45\pi}{\alpha}} \left(\frac{B_{\text{crit}}}{B} \right) \omega_{\text{pl}} \simeq 0.16 \sqrt{n_{21}} \left(\frac{B_{\text{crit}}}{B} \right) \text{ keV}. \quad (5)$$

Alternatively, near the cyclotron frequency, $\omega_{\text{cyc}} = m_e(B/B_{\text{crit}})$, the ratio $(\varepsilon/v)^6$ is, in vacuum,

$$\left(\frac{\varepsilon}{v} \right) = \frac{\alpha}{180\pi^2} \frac{m_e^3}{e^2} \frac{(B/B_{\text{crit}})^4}{n_e} \simeq 2.48 \times 10^5 \frac{(B/B_{\text{crit}})^4}{n_{e21}}, \quad (6)$$

so that for $B/B_{\text{crit}} \in (0.04 - \frac{1}{4})$, $(\varepsilon/v) \in (6 - 10^3) \times n_{e21}^{-1}$.

The use of the magnetized vacuum modes may be thus justified by suitably constraining the optical depths in the direction of preferential photon escape ($\hat{\mathbf{l}}_z$ for collisionally stopped accretion and $\hat{\mathbf{l}}_\rho$ for accretion columns). A maximum value of $\tau_c = 3 \times 10^{-3}$ easily permits the use of the magnetic modes in either geometry, since the scale heights in slabs are of order of 100 cm (§ 3.1), and cylinders have a much larger length scale in the escaping direction $\sim (A_{\text{cap}}/\pi)^{1/2}$, implying still lower electron number densities.

Photon mode-dependent radiative transfer will not be considered here, and dispersive effects will thus be ignored. The proper photon polarization vectors to be utilized in the calculation of magnetic Compton scattering (§ 4) are those that diagonalize the photon's polarization tensor (Shabad 1976),

$$\epsilon_v^j = \frac{b_v^j}{\sqrt{-b_\mu b^\mu}}, \quad \text{with} \quad b_\mu^\perp = F_\mu^\nu k_\nu \quad \text{and} \quad b_\mu^\parallel = F_\mu^{\star\nu} k_\nu. \quad (7)$$

For a magnetic field in the $\hat{\mathbf{l}}_z$ direction, the modes become

$$\epsilon_v^\perp = \frac{\mathbf{k} \times \mathbf{B}}{|\mathbf{k} \times \mathbf{B}|} \quad \text{and} \quad \epsilon_v^\parallel = \frac{\mathbf{k} \times (\mathbf{k} \times \mathbf{B})}{|\mathbf{k} \times (\mathbf{k} \times \mathbf{B})|} \quad (8)$$

by making use of gauge invariance to rewrite the parallel mode, $b^{\mu\parallel} \rightarrow b^{\mu\parallel} k_0 - k_3 k^\mu$.

The normal modes above, equation (8), will be referred to as the parallel and perpendicular mode. They provide a suitable basis for the calculation of the scattering cross section in § 4.

3.3.2. Electron Spin States

In the quantization of the Dirac field, the Dirac-Landau operator (e.g., Melrose & Parle 1983),

$$[-i\gamma_\nu(\partial^\nu - eA^\nu) - m], \quad \Psi(\mathbf{x}) = 0, \quad (9)$$

has degenerate electron wave functions because the Hamiltonian derived from this operator is not explicitly dependent on a relativistic spin operator.

⁶ Omitting the plasma component singular behavior at resonance.

The spin degeneracy is removed by requiring that the spinors be eigenfunctions of a particular choice of relativistic spin operator with highly desirable symmetry properties:

$$\mu \equiv m\{\boldsymbol{\sigma} + \rho_y \boldsymbol{\sigma} \times [\hat{\mathbf{p}} + e\mathbf{A}(x)]\}. \quad (10)$$

States of this Sokolov-Ternov (1968) *magnetic moment operator* commute with the radiative corrections to the Dirac-Landau equation (Herold, Ruder, & Wunner 1982) and provide for a proper basis to construct dressed electron propagators that motivate the natural line widths in the magnetic Compton-scattering cross section (Graziani 1993).

The perpendicular component of the momentum for fermion particles is discretely quantized because of the non-commutativity of the *physical* momentum operators in the plane perpendicular to the magnetic field. The corresponding eigenvalues are given by the roots of

$$2neB = p_n^2 \equiv E^2 - m^2 - p_z^2. \quad (A7)$$

The quantization of the relativistic energy of electrons thus yields resonances in the photon-electron interactions. Ions and other hadrons have similar resonances, but they occur at lower energies because of their higher masses. These resonances are thus inconsequential in the treatment of cyclotron line production at X-ray energies.

4. THE CROSS SECTION FOR MAGNETIC COMPTON SCATTERING

4.1. The First-Order Processes

Although the *second-order* cyclotron scattering process involves a sum over virtual intermediate states, it may be *approximately* interpreted as *first-order* cyclotron absorption immediately followed by emission (see eqs. [11], [15], and [57]). For later reference we collect the first-order processes' formulae below.

4.1.1. Cyclotron Absorption

The *rest frame* relativistic absorption cross section (primed quantities), summed over final spin states and averaged over initial photon polarization, is given by (Daugherty & Ventura 1978; Harding & Daugherty 1991)

$$\sigma_n^{\text{abs}}(\theta') = \frac{\alpha\pi^2}{(n-1)!} \frac{1}{E_n'} \delta(E_n' - k'^0 - 1) eZ^{n-1} - Z \times \left[(1 + \cos^2 \theta') + \frac{Z}{n} \sin^2 \theta' \right], \quad (11)$$

where k'^ν and θ' refer to the incident photon,

$$Z \equiv \frac{(k'^0 \sin \theta')^2}{2B'} \quad (12)$$

characterizes the relative photon strength for excitation of perpendicular momentum quanta,

$$E_n' = (1 + q'^2 + 2nB')^{1/2}, \quad \text{with} \quad B' \equiv \left(\frac{B}{B_{\text{crit}}} \right), \quad (13)$$

is the energy of the final excited electron state (or that of the resonant intermediate state for the corresponding scattering process), and where $n(=n_{\text{int}})$ will be understood to be the Landau level for such state in what follows. Furthermore, strict energy conservation (i.e., with *zero line width*) requires

that the incident photon energies, k'^0 , obey

$$k'^0 = \omega_n^{\text{res}} = \frac{(1 + 2nB' \sin^2 \theta) - 1}{\sin^2 \theta}. \quad (14)$$

4.1.2. Cyclotron Emission

In a frame in which $p_z = 0^7$, the differential transition rate for cyclotron emission from an excited Landau level n to the ground state, averaged over initial and summed over final spin states, is (Latal 1986)

$$\frac{d\Gamma_{n \rightarrow 0}(\mu)}{d\Omega} = \frac{\alpha}{2\pi} \frac{1}{(n-1)!} \frac{1}{E_n^2} \frac{2nB'}{E_n} e^{-Z} Z^{n-1} \times \frac{1}{(1+\xi)^2} \left[1 + 2nB' - \frac{1}{\xi} \right], \quad (15)$$

where

$$\begin{aligned} d\Omega &= 2\pi d\mu, \quad \nu \equiv 1 - \mu^2, \\ E_n &= (1 + 2nB')^{1/2}, \quad \xi \equiv (1 - 2nB'\nu^2/E^2), \\ Z &= (k^0)^2(1 - \mu^2)/2B' = n(1 - \xi)/(1 + \xi). \end{aligned}$$

4.2. The Scattering Cross Section

Our choices of photon polarization and electron spin states were motivated in §§ 3.3.2 and 3.3.1. With this choice of basis states, the cross section has been computed recently by Sina (1996; see also the Appendix).

With the following definitions:

$$Z_{c1}(n) \equiv \sum_{s_n} \left[\frac{D_{n_f,n}^{(++)}(k_f) H_{n,n_i}^{(++)}(k_i)}{k_i^0 + E_i - (E_n^{(1)} - i\Gamma_n/2)} \right] + \left[\frac{D_{n_f,n}^{(+-)}(k_f) H_{n,n_i}^{(+-)}(k_i)}{k_i^0 + E_i + (E_n^{(1)} - i\Gamma_n/2)} \right], \quad (16)$$

$$Z_{c2}(n) \equiv \sum_{s_n} \left[\frac{H_{n_f,n}^{(++)}(k_i) D_{n,n_i}^{(++)}(k_f)}{-k_f^0 + E_i - (E_n^{(2)} - i\Gamma_n/2)} \right] + \left[\frac{H_{n_f,n}^{(+-)}(k_i) D_{n,n_i}^{(+-)}(k_f)}{-k_f^0 + E_i + (E_n^{(2)} - i\Gamma_n/2)} \right]; \quad (17)$$

the general expression for the differential cross section is

$$\begin{aligned} \frac{d\sigma}{d\Omega_f} &= \frac{\alpha^2}{(1 - \beta_i \cos \theta_i)(1 - \beta_f \cos \theta_f)} \frac{k_f^0}{k_i^0} \sum_{n=0}^{\infty} \\ &\times \left\{ |Z_{c1}(n)|^2 + |Z_{c2}(n)|^2 \right. \\ &\times \left(\sum_{m=0}^{\infty} [Z_{c1}(n)^* Z_{c2}(m) + Z_{c1}(n) Z_{c2}(m)^*] \right) \\ &\times J_{m+n-n_i-n_f}(-eBk_{\perp i} k_{\perp f}) \end{aligned} \quad (18)$$

where the vertex functions: $D_{n,n_i}^{(\xi\eta)}(\mathbf{k})$ and $H_{n,n_i}^{(\xi\eta)}(\mathbf{k})$ follow from Daugherty & Bussard (1980) with the notation of the Appendix.

4.3. Radiative Line Widths

Line widths in the absorption cross section are traditionally put in “by hand” through a substitution of the energy conserving δ -function by a Lorentzian function motivated by an analogy to the radiation damping lifetime of an excited state in nonrelativistic quantum mechanics.

⁷ Note, however, that this 0p-frame is *not* the “rest frame” (see § 5.6).

For the scattering cross section, on the other hand, a *complexification* of the virtual particle energy,⁸ with the imaginary part being equal to the decay rate of the resonant state $E_n \rightarrow E_n - i\Gamma_{n,s}/2$ (see eq. [19]), leads to a relativistic quasi-Lorentzian *resonant* profile in the cross section (Harding & Daughterty 1991) consistent at nonrelativistic energies with the Lorentzian prescription (see the discussion in § 5).

We will adopt this prescription, noting that it is on firm physical ground provided that use is made of the proper electron states (e.g., Graziani, Harding, & Sina 1995) and as long as the resonant part of the cross section dominates its behavior.

The radiative widths, $\Gamma_{n,s}$, are thus derived from the relativistic cyclotron transition rates of Sokolov & Ternov (1968; see also Harding & Preece 1987) and are summed over final electron and photon states,

$$\Gamma_{n,s} = \sum_{n'=0}^{n-1} \sum_{s'=0}^1 \Gamma_{n',s'}^{\prime\prime}. \quad (19)$$

5. THE MONTE CARLO PROGRAM: A PHOTON'S HISTORY

We now follow the history of a typical X-ray photon as it propagates through an electronic atmosphere permeated by a uniform magnetic field.

It will be evident that exact evaluation of the probability distributions involved in the Monte Carlo sampling is often computationally impractical and that some approximations must be invoked. Careful consideration of the range of validity of these approximations and possible sources of inaccuracy are examined.

5.1. Photon Injection

Let n_ω be a prescribed *incident* photon number density (with $\omega = 2\pi\nu$ and I_ν the radiation intensity):

$$n_\omega d\omega \equiv \left(\frac{1}{hc}\right) \frac{I_\nu}{\nu} d\nu = \left[\int \frac{d\mu d^2N_\omega}{(d\omega d\mu)} \right] d\omega, \quad (20)$$

and note that the cyclotron spectra are formed by unbiased injection of individual photons from N_E identical energy bands evenly spread throughout a spectral range.

From standard Monte Carlo methodology, the probability that any one injected photon will belong to the i th energy interval is equal to the normalized partial integral of n_ω over the corresponding energy bin. Keeping the total number of photons shone on the layer, N_{tot} , constant, each energy bin would contribute with a number of photons dictated by this probability *if* the injection were *biased*. For *unbiased* injection, however, each energy bin contributes with an equal number of photons, but the injected photon and all its spawned processes carry a spectral weight equal to the probability corresponding to the photon's *injection* bandwidth.

For the hard continuum that is the object of this investigation, we have a power-law continuum with $\alpha = -1$, so that $n_\omega \propto \omega^{-1}$ and the primitive of n_ω is proportional to $\ln(\omega)$. Mapping the probability distribution onto the random number domain yields

$$\omega(\text{Rn}) = \omega_{\min} \left(\frac{\omega_{\max}}{\omega_{\min}} \right)^{\text{Rn}}, \quad (21)$$

⁸ Or rather, the introduction of an imaginary part in the pole of the electron propagator (Graziani 1993).

where Rn is a random number. The probability weights for unbiased injection may be readily evaluated using this relation.

5.2. Photon Propagation

5.2.1. The Mean Free Path Length

The injected photon, k^ν , will now propagate through the electronic medium, and an assessment of its travel path before interaction with the medium is required.

It is convenient to introduce continuum normalizations for the following functions of k^ν : the differential cross section $d^2\sigma/(d\Omega_f dk^0)$, the total profile $\mathcal{P}_{\text{sc}}$, the mean free path λ_{sc} , and the optical depth τ_{sc} . The continuum quantities are characterized by the Thomson cross section σ_{Th} . Moreover, it will be understood that the only interaction considered is magnetic Compton scattering, and the subscripts are conveniently dropped.

With these definitions, the mean free path length for isotropic and isothermal media (see Araya 1996 for a more general formulation) may be written as

$$\lambda(k^0, \mu) = \frac{1}{\langle \sigma(k^0, \mu) \rangle}, \quad (22)$$

where σ_{sc} is averaged over a choice of electron distribution function, $f_e(p, T)$ see equation (23), and the dependence of τ on column density, N_l , is thus shifted to the Thomson optical depth of the scattering layer, τ_c .

5.2.2. Thermal Averaging of the Cross Section

The magnetic Compton-scattering cross section used above is the one given by equation (18), summed over final photon states (energy, solid angle $[d^3k]$ and polarization) and averaged over initial photon polarization. The average of $\sigma(k^\nu)$ is taken over a one-dimensional relativistic electron momentum distribution $f_e(p)$. By a well-justified assumption (§ 3.2.1), the electrons are initially in the ground Landau level, and the plasma's thermodynamic state is determined solely by the so-called parallel (to the B field) temperature. Under this circumstance, the probability distribution for relativistic one-dimensional thermal electrons is given by (with $k_B = 1$)

$$f_e(p) dp = N_{T_e} \exp \left\{ -\frac{\sqrt{1 + p_e^2} - 1}{T_e} \right\} dp, \quad (23)$$

where T_e is the parallel electron temperature and N_{T_e} the normalization.

The cross section in the lab frame is given by

$$\sigma_{\text{lab}} = \sigma_{\text{rf}}(1 - \mu\beta), \quad (24)$$

where $\mu = \hat{k} \cdot \hat{B}$ is the cosine of the propagation angle with respect to the magnetic field, and the factor $(1 - \mu\beta)$ can be thought of as a “relative velocity” involving massless particles (Sina 1996).

The equation for the total profile in homogeneous media follows:

$$\mathcal{P} = \langle \sigma(k^0, \mu) \rangle_{p_e} = \int_{-\infty}^{\infty} dp f_e(p) (1 - \mu\beta) \sigma_{\text{rf}}(k^0, \mu'). \quad (25)$$

5.2.3. The Photon's Travel Path

The mean free path $\tilde{\lambda}(k)$ is the statistical mean of the spatial length that a photon, k^ν , will traverse. The *normal-*

ized optical depth τ (normalized to τ_c) is in fact identical to the inverse dimensionless path λ . The probability that the photon will propagate an optical path $\delta\tau_c$ before scattering in a medium having mean free path $\bar{\lambda}$ is

$$\mathcal{P}r_{\text{move}}(\delta\tau_c) = \exp \left\{ -\frac{\delta\tau_c}{\bar{\lambda}} \right\}, \quad (26)$$

which is a function of $\delta\tau$ with identical primitive except for an inconsequential scale. The normalized primitive can be easily inverted into

$$\delta\tau_c = -\bar{\lambda} \ln(Rn) \quad (27)$$

so that the propagation path, from injection as well as between scattering events, may be readily sampled stochastically.

5.3. Electron Parallel Momenta

The test photon will now move a distance $\delta l \propto \delta\tau_{\text{Th}}$ in accordance with equations (26) and (27), and at the end of this journey it will scatter with an electron. The electron will be initially in the ground Landau level, and its relevant quantum state is determined solely by its parallel momentum (the only spin allowed is $s = -$ (down), and the x coordinate of its orbit is inconsequential).

With primes denoting *rest frame* evaluation, the differential probability⁹, $\mathcal{A}rg(p, k^0, \mu)\delta p$, for a photon with lab frame k^0 and μ to scatter off an electron with parallel momentum $\in [p, p + \delta p]$ is

$$\mathcal{A}rg(p, k^0, \mu)\delta p \equiv \frac{1}{\mathcal{P}} f_e(p)(1 - \mu\beta)\sigma_{rf}(k^0, \mu')\delta p, \quad (28)$$

with $\mathcal{P}$ as given by equation (25).

The stochastic sampling of the target electron's momentum, q , may in principle be achieved by matching a random number to the partial integral of $\mathcal{A}rg(p, k^0, \mu)\delta p$ from $p_{\min}$ up to $q = p_i$ (i.e., the initial momentum before scattering):

$$Rn \leftrightarrow \int_{p_{\min}}^{p_i} dp \mathcal{A}rg(p, k^0, \mu), \quad (29)$$

where the maximum allowed momenta, $p_{\min}$ and $p_{\max}$, are numerically set to $\pm 45kT_e$, and the evaluation of σ in the initial electron's rest frame facilitates our discussion below.

Numerical results show that the Monte Carlo line features are extremely sensitive to the sampling of electron momenta. However, repeated numerical evaluation of the partial integral involved for each scattering event is computationally costly because of the sharply peaked nature of the cross section when k'^0 nears any resonance. Worse still, the form of $d^2\sigma_{\text{sc}}/(d\Omega dk^0)$ in § 4 does not lend itself to straightforward analytical evaluation.

Two methods are developed to sample the scattering electron's momenta: (1) a *linear* three-dimensional invertible interpolation scheme in a quadrature sum of $\mathcal{A}rg(\delta p, \delta k^0, \delta\mu)$ utilizing the exact form for magnetic Compton scattering equation (18), and (2) a “half-interval” searching method using a semianalytical approximation to the integral allowed by virtue of the simpler form for the absorption

cross section. In either case, extensive study of the integrand's behavior throughout the interval of integration is essential.

5.3.1. Absorption versus Scattering

The *absorption* cross section, equation (11), provides most of the qualitative behavior of the scattering cross section without the obscure mathematical complexity that the latter pertains. Let us compare their resonant (singular) behavior.

For cyclotron absorption, substitution of the energy conserving δ -function in equation (11) by a normalized Lorentz profile may be justified in the nonrelativistic limit (in fact, relativistically the absorption process must be thought of as a scattering event that leaves the electron in the excited resonant state). The absorption cross section is then proportional to the sum of two Lorentzian profiles corresponding to the radiative decay from up and down spin states (see eq. [19]).

For a given resonance, n , the relative strength of the two transitions is fixed by requiring that at resonance both Lorentzian functions contribute equally to the cross section in the nonrelativistic limit. Thus $\sigma_d/\sigma_u = \Gamma_u/\Gamma_d$, and the singular part of the cross section takes on the form (while suppressing primes)

$$\delta(E_n - k^0 - 1) \rightarrow \mathcal{L}_d(k^0, \mu),$$

where

$$\mathcal{L}(k^0, \mu) \equiv \frac{1}{\pi(1 + \Gamma_u/\Gamma_d)} \left[L_d(k^0, \mu) + \frac{\Gamma_u}{\Gamma_d} L_u(k^0, \mu) \right], \quad (30)$$

with

$$L_j(k^0, \mu) \equiv \left(\frac{(2/\Gamma_j)}{\{(2/\Gamma_j)[k^0 - \omega_n(\mu)]\}^2 + 1} \right), \quad j = u, d,$$

and where the Landau level subscript n in the decay rates is suppressed.

In the relativistic regime this nonrelativistic Lorentz function, $L_j(k^0, \mu)$, should be replaced by relativistically modified Lorentzian radiative line profiles, $\tilde{L}_j(k^0, \mu)$, as derived from the scattering cross section. Following Harding & Daugherty 1991, with

$$\tilde{L}_j(k^0, \mu) \equiv \left(\frac{(2/\Gamma_j)}{\{[2/(\Gamma_j E_n)][k^0 - \omega_n(\mu)]\}^2 + 1} \right), \quad (31)$$

let us define

$$\mathcal{K}(k^0, \mu) \equiv \frac{1}{\pi(1 + \Gamma_u/\Gamma_d)} \left\{ \tilde{L}_d(k^0, \mu) + \frac{\Gamma_u}{\Gamma_d} \tilde{L}_u(k^0, \mu) \right\}, \quad (32)$$

which, in the nonrelativistic limit $E_n \rightarrow 1$, behaves as expected

$$\mathcal{K}(k^0, \mu) \rightarrow \mathcal{L}(k^0, \mu).$$

With $\mathcal{L}(k^0, \mu')$ as defined in equation (30), $\mathcal{K}(k^0, \mu')$ as defined above and with primed quantities indicating *rest frame* evaluation,

$$\frac{\sigma_{\text{sc}}}{\sigma_{\text{abs}}} \simeq \frac{\mathcal{K}(k^0, \mu')}{\mathcal{L}(k^0, \mu')}. \quad (33)$$

⁹ Note the change in notation for the differential probability,

$$\mathcal{P}r(p, k^0, \mu)\delta p \rightarrow \mathcal{A}rg(p, k^0, \mu)\delta p.$$

$\mathcal{A}rg(p, k^0, \mu)$ here merely stands for the integrand in eq. (25).

The relation is approximate because off-resonance terms are ignored, and only the quadratic (in Γ) resonant terms are kept to obtain equation (32).

To sample electron momenta analytically, use of the relativistically modified Lorentzian function, equation (32), is complicated by the dependence of $\tilde{L}$ on k'^0 (e.g., $E' = E'(q' = k'^0 \sin \theta)$ in eq. [13]), and this precludes the ease of solution that the Lorentzian profile provides. More importantly, since the Lorentz function, L , has a well-defined analytical primitive (i.e., the *arctangent*), its use considerably eases the evaluation of the electron momentum probability distribution, and facilitates the qualitative analysis of the integrand $\mathcal{A}rg(p, k^0, \mu)$ for the more general case of relativistic scattering.

5.3.2. The Integrand's Analytical Behavior

Although the integration in equation (29) is performed over parallel momentum, the cross section is a function of k'^0 and μ' , which are the Lorentz-boosted photon quantities evaluated in the rest frame of the initial electron.

In the absorption approximation, the δ -function for k'^0 may be written as

$$\delta[k'^0 - \omega_n(\mu')] = \sum_{i=+,-} \left| \frac{1}{\partial_p g} \right|_{P_i} \delta(p - P_i), \quad (34)$$

where

$$g(p, k^0, \mu) = k'^0 - \omega_n^{\text{res}}(\mu'), \quad (35)$$

and where $P_{\pm}$ are the zero line width solutions to the resonant condition

$$k'^0 = \omega_n^{\text{res}}[\mu'(p)], \quad (36)$$

which must be found numerically (Araya 1996).

For a photon at resonance, $k'^0 = \omega_n[\mu'(p)]$, and given k^0 and the Lorentz boost along $\hat{1}_z$,

$$k'^0 = k^0 \gamma(1 - \beta\mu) \quad \text{and} \quad \mu' = \frac{\mu - \beta}{1 - \beta\mu}, \quad (37)$$

the zero line width solutions can be written as

$$P_{\pm}^{\text{zlw}} = \frac{1}{1 - \mu^2} (\mu b_r \pm \sqrt{b_r^2 + \mu^2 - 1}), \quad (38)$$

where

$$b_r \equiv \frac{\omega_n^{\text{res}}(\mu')}{k^0} = \gamma(1 - \beta\mu)$$

is an implicit function of p (see eq. [37]).

For k'^0 values (or p values) not necessarily at resonance, the function

$$b(p) \equiv k'^0/k^0 = \sqrt{1 + p^2} - p\mu \quad (39)$$

has a minimum at $p = \cot \theta$ and the asymptotic behavior:

$$b(p) \rightarrow (1 \pm \mu)p, \quad \text{as} \quad p \rightarrow \pm \infty.$$

If $b_{\min} \ll b_r = \omega_n(\mu')/k^0$, this implies that two P^{zlw} solutions exist, since the resonant condition $k'^0 = \omega_n(\mu')$ is met twice in the whole range of p values. Now,

$$b_{\min} = |b(p)|_{p=\cot \theta} = \sin \theta,$$

and this is the significance of the discriminant in equation (38).

Conversely, if $b_{\min} \simeq b_r + \epsilon$, there is one P^{zlw} solution if $b_r = \sin \theta$ or none if $b_r < \sin \theta$. In the latter case, the highest value that the Lorentzian representation of the δ -function attains is

$$\mathcal{L}[k'^0 = k^0 \sin \theta, \mu' = \mu'(p = \cot \theta)] \equiv \mathcal{L}_b^{\text{max}},$$

where the subscript b denotes “beyond the cutoff energy.” Next, note that

$$\mu'(\mu, p = \cot \theta) = \left| \frac{\mu - \beta}{1 - \beta\mu} \right|_{\beta=\mu} = 0, \quad (40)$$

so that $\mathcal{L}_b^{\text{max}} = \mathcal{L}(k^0 \sin \theta, 0)$.

Lastly, the case of one P^{zlw} also has $p = \cot \theta$ and corresponds to the relativistic cut off energy

$$\omega_{\text{Cut}} = \frac{\sqrt{1 + 2B'} - 1}{\sin \theta}. \quad (41)$$

5.3.3. A Semianalytical Approximation to the Profile Partial Integrals

In order to treat the integral in equation (29) semi-analytically, a linear transformation of the (the rest frame) energy δ -function into momentum space is highly desirable. To justify the use of equation (42) below, a linear map between k^0 and p_e spanning at least a few energy line widths around resonance must be found:

$$k'^0 = \omega_n + \Gamma_{u,d}/2 \rightarrow p = P_i^{\text{zlw}} \pm \Delta_{u,d}^i/2,$$

where the $\pm$ ambiguity arises because an increment in energy from resonance $\delta k'^0$ may correspond either to a increment or a decrement in momentum $\rightarrow \pm \delta p$ at constant incident lab frame photon energy k^0 .

Under the above premise, and since the radiative line widths fulfill

$$\Gamma_{u,d} \ll k'^0 - \omega_n^{\text{res}}(\mu') (\equiv \delta\omega'),$$

for most values of p_e , one may again utilize the sum of spin-dependent Lorentz functions as a p -space representation for the δ -function.¹⁰ Careful analysis of the integrand's behavior is needed to establish when the δ_p -function representation is justified.

Following equation (30) and noting that the ratio of line widths in p space obey $\Delta_u/\Delta_d = \Gamma_u/\Gamma_d$ if the relation between k^0 and p_e is linear (see eq. [43]), one may use

$$\delta(p - P_i) \rightarrow \tilde{\mathcal{L}}^i(p - P_i),$$

where

$$\tilde{\mathcal{L}}^i(p - P_i) \equiv \frac{1}{\pi(1 + \Delta_u/\Delta_d)} \left[\tilde{L}_d^i(k^0, \mu) + \frac{\Delta_u}{\Delta_d} \tilde{L}_u^i(k^0, \mu) \right], \quad (42)$$

with

$$\tilde{L}_j(k^0, \mu) \equiv \left\{ \frac{2/\Delta_j^i}{[(2/\Delta_j^i)(p - P_i)]^2 + 1} \right\}_{i=+,-}^{j=u,d},$$

¹⁰ Alternatively, one may treat each Lorentzian profile, L , in eq. (30) as a δ -function in k^0 space, transform them to p -space with eqs. (34) and (35), and use a Lorentzian representation of the δ -function in momentum space p_e .

and where $\Delta_{u,d}^{+,-}$ is the spin state dependent (subscript) width of the profile in p -space around either zero line width (superscript) solution P_i .

Near the rest frame resonant energy mapped onto one of the zero line width P solutions, P_i^{zlw} , the desired linear relation between Γ and Δ can be easily extracted from an analysis of the variation of δp with $\delta b(p)$ as defined in equation (39). The result for the “line widths” in p space is (Araya 1996)

$$\Delta_{u,d}^{\pm} = \frac{\Gamma_{u,d}}{k^0 \sin^2 \theta} \left\{ \frac{b_r}{\sqrt{b_r - \sin^2 \theta}} \pm \mu \right\}. \quad (43)$$

Now, the relative position of the P_{zlw} 's with respect to $p = 0$ and each other determines the behavior that the integrand, $\mathcal{A}rg(p, k^0, \mu)$, takes throughout the p range. Consider the case where $P_+^{\text{zlw}} - P_-^{\text{zlw}} \gg \Delta_d$ ($\Delta_d > \Delta_u$ following the trend in energy space: $\Gamma_d > \Gamma_u$). By direct examination of the integrand,

$$\mathcal{A}rg(p, k^0, \mu) dp = \frac{1}{\mathcal{P}} f_e(p) (1 - \mu\beta) \sigma_{r,f}(k^0, \mu') dp,$$

and by assuming δ -function-like behavior for the cross section, it follows that the P_{zlw} solution closest to 0 will dominate the integral. Unless one of the $P_{\text{zlw}} \rightarrow 0$ or both $|P_{\text{zlw}}| \rightarrow \infty$, the contribution from the electron distribution function will be negligible. In general, however, one may consider three separate contributions:

$$\begin{aligned} \frac{1}{\mathcal{P}} \int_{p_{\min}}^q dp \mathcal{A}rg(p, k^0, \mu) \\ \simeq \frac{1}{\mathcal{P}} \left\{ \left| \sigma(k^0, \mu) \right|_{p=0} \int_{p_{\min}}^q dp f_e(p) \right\} \\ + \left\{ \sum_{i=-,+} \left| \frac{f(p)}{g'(p)} \right|_{p_i} (1 - \beta_i \mu) \int_{-\infty}^q dp \tilde{\mathcal{Z}}(p - P_i) \right\}, \quad (44) \end{aligned}$$

where $\tilde{\mathcal{Z}}(p - P_i)$ is given by equation (42),

$$g(p) = [k'^0 - \omega_n(\mu')] \quad \text{and} \quad g'(p) = \partial_p g(p).$$

The normalization factor or total profile, $\mathcal{P}$, is given by the sum of all the factors in front of the integrals, since each separate integral is normalized. Likewise, the relative contribution from each part is given by the ratio of the respective factor to the sum.

A *semianalytical* inversion of equation (29) for stochastic sampling of electron's parallel momenta is achieved as follows: the zero line width solutions are found numerically, as previously indicated, and the partial integral from the first term in the right-hand side of equation (44) may be easily evaluated, stored, and searched numerically. Furthermore, the Lorentzian profiles are simple to integrate and evaluate (see eq. [42]), and the relative strengths of the first term and of the P_{zlw} 's contributions are easily computed from equations (11), (35), and (23). A “half-interval” searching method using this semianalytical approximation to the integral rapidly converges to the target electron's momentum. This is, in essence, the semianalytical scheme used in the Monte Carlo code.

Equation (43), however, is invalid for energies approaching the relativistic cutoff energy

$$k'^0 = k^0 \sin \theta. \quad (45)$$

Consider the following two permissible cases near the cutoff energy (with $\delta b_r > 0$ and $\varepsilon \approx \mathcal{O}(2b_r \delta b_r) > 0$):

$$\begin{aligned} \varepsilon &\equiv b_r^2 - \sin^2 \theta \rightarrow +0, \\ -\varepsilon &\equiv b_r^2 - \sin^2 \theta \rightarrow -0. \end{aligned} \quad (46)$$

If an attempt is made to find a relation between k'^0 and p_e in either case, it will become obvious that the relation is no longer linear (Araya 1996).

Noteworthy, in the second case, possible electron momenta p_e appear beyond that corresponding to the relativistic cutoff energy:

$$p_{\text{Cut}} \equiv p(b_r = \sin \theta) = \cot \theta \quad (47)$$

because of the finite energy line width.

5.3.4. Numerical Approaches

The analysis of the integrand performed in the preceding section greatly aids the construction of a numerical interpolation scheme that constitutes the alternative to the semi-analytical method for choosing the electron's momentum. On the other hand, a “brute force” method utilized occasionally (see below) during run time evaluates the partial integral repeatedly and, through splitting of the integration range, methodically searches for the partial integral that matches a generated random number down to some given tolerance. Our MCP makes use of an adaptive recursive Runge-Kutta routine (see Numerical Recipes; Press et al. 1988), which proved to be stable most of the time but also computationally quite expensive because of the singular nature of the integrand. This routine used the exact form for the scattering cross section and was utilized to create the tables of partial integrals used in the interpolation scheme.

The numerical interpolation scheme must also be sensitive to the singular nature of the integrands involved. Additionally, it must be invertible. Any procedure other than linear interpolation adds significant computational complexity to the inversion procedure. Thus with the simple and useful employment of logarithmic derivatives, an uncomplicated three-dimensional (i.e., in μ_i , k_i^0 and p_i) interpolation routine is the choice made for finding the scattering electron's momentum *except* for the last scattering. The last scattering dictates that the choice of p be implemented much more accurately. The semianalytical method underlined above is then utilized if the energy of the incident photon is far from the relativistic cutoff energy. Otherwise, the brute force method must be employed (note that even this method is not 100% stable).

The use of a linear interpolation, even when aided by logarithmic derivatives, is not sufficiently reliable to sample the parallel momentum distribution. One has to resort to a rather elaborate scheme to place the sampling grid points strategically and thus minimize irregular interpolations. Some facts regarding the integrand's analytic behavior discussed in the preceding section are tremendously useful in the placing of the grid points (Araya 1996).

5.4. The Kinematics of Magnetic Compton Scattering

Given the initial electron parallel momentum, p , the initial and final electron Landau levels, n and m , and the outgoing and incoming photon's angle with respect to the $\mathbf{B}$ -field, $\mu_{i,f} = \hat{\mathbf{B}} \cdot \hat{\mathbf{k}}_{i,f}$, the outgoing photon energy, k^0 , may be trivially computed.

Energy and parallel momentum conservation yield

$$k_i^0 + p^0 = k_f^0 + q^0 \quad \text{and} \quad k_i^0 \mu_i + p = k_f^0 \mu_f + q, \quad (48)$$

where q is the final electron momentum.

Utilizing equation (13) for q^0 and p^0 and eliminating q^2 , a standard quadratic equation for k_f^0 follows with coefficients

$$a = 1 - \mu_f^2, \quad -\frac{b}{2} = [k_i^0(1 - \mu_i \mu_f) + (p^0 - p \mu_f)],$$

$$c = (k_i^0)^2(1 - \mu_i^2) + 2k_i^0(p^0 - p \mu_i) + 2B'(n - m), \quad (49)$$

and discriminant

$$\frac{b^2}{4} - ac = [k_i^0(\mu_f - \mu_i)]^2 + (p^0 - p \mu_f)^2$$

$$+ 2k_i^0(\mu_f - \mu_i) - 2(1 - \mu_f^2)B'(n - m).$$

5.5. Electron Excitation and Photon Spawning

Originally the electron is in the ground Landau level, $n_i = 0$ (§ 3.2.1), and we choose to treat the scattering process in the rest frame of the initial electron, performing the required Lorentz transformations to and from this reference frame. If the electron is left in an excited state on scattering, $n_f \neq 0$, then one has to consider subsequent de-excitation of this state, thus giving birth to *spawned photons* with energies corresponding to the anharmonic spacing between the excited Landau level and the ground state (e.g., first-order cyclotron emission).

Let us denote the cross section for a photon near the n th resonance (i.e., $k'^0 \approx nB'$) to leave an electron with $p_i = 0$ and $n_i = 0$ in the $n_f = l$ excited level by $\mathcal{S}_l(k'^0, \mu')$. In the initial electron's rest frame (primed quantities) the expression for the total scattering cross section is found from equation (18) by summing over all possible (discrete) final electron states (see § 4):

$$\sigma_n(k'^0, \mu') = \sum_{l=0}^{\infty} \int d\Omega_f \frac{d\sigma}{d\Omega_f} \equiv \sum_{l=0}^n \mathcal{S}_l(k'^0, \mu'), \quad (50)$$

where the label n is superfluous (it just makes explicit that the photon's rest frame energy is near the n th resonance) and the integral over the final photon states has been explicitly performed. Note that in this frame the sum is bounded by the maximum energetically accessible level for such photon.

The (discrete) probability for the photon to leave the electron in excited state $m \leq n$ on scattering is simply the ratio of the scattering cross section into Landau state m to the total scattering cross section into all energetically allowed final Landau levels:

$$\mathcal{P}r_n(0 \rightarrow m) = \frac{\mathcal{S}_m(k'^0, \mu')}{\sigma_n(k'^0, \mu')}. \quad (51)$$

With these discrete probabilities, the stochastic implementation of this process is straightforward.

$$\text{Rn} \leftrightarrow \{\mathcal{D}_i, i = 1, \dots, n\},$$

where

$$\mathcal{D}_i = \left[\frac{\sum_{l=0}^{i-1} \mathcal{S}_l(k'^0, \mu')}{\sigma_n(k'^0, \mu')}, \frac{\sum_{l=0}^i \mathcal{S}_l(k'^0, \mu')}{\sigma_n(k'^0, \mu')} \right]_i,$$

so that given a random number, Rn , one finds the corresponding domain $\mathcal{D}_m$ to place the electron in the m th Landau level on the scattering event.

If $m \neq 0$, the electron will radiatively de-excite (see § 3), producing “nearly” resonant spawned photons (see eq. [1]). In the electron rest frame, the probability for the electron to de-excite from level m to level m' is equal to the relativistic cyclotron transition rate into level m' summed over final electron spin states,

$$\Gamma_{m'}^m = \sum_{s,s'=u,d} \Gamma_{m',s'}^{m,s}, \quad (52)$$

normalized to the total rate from the m level:

$$\mathcal{P}r(m \rightarrow m') = \frac{\Gamma_{m'}^m}{\sum_n \Gamma_n^m}. \quad (53)$$

Again, with the domains defined as

$$\mathcal{D}_i = \left[\frac{\sum_{l=0}^{i-1} \Gamma_l^m}{\sum_{l=0}^n \Gamma_l^m}, \frac{\sum_{l=0}^i \Gamma_l^m}{\sum_{l=0}^n \Gamma_l^m} \right]_i, \quad \text{for } i = 1, \dots, n; \quad (54)$$

the choice of final Landau level on cyclotron emission is made just as in the scattering case above.

In § 5.4, it was found that given the set of quantities: k_i^0 , p_i , n_i , n_f , μ_i , and μ_f , the values of p_f and k_f^0 trivially follow. Thus to complete the scattering process, the angle of the outgoing photon needs to be sampled next.

5.6. Angular Redistribution: the On-Shell Approximation

Sampling the angular redistribution probability for resonant photons through the use of the exact second-order scattering cross section (eq. [18]) is cumbersome and numerically impractical. On the other hand, an excellent approximation near resonance may be obtained by making use of the simpler expression for cyclotron emission. These rates, summed over final electron states, have a very similar functional form to those of cyclotron absorption (compare eqs. [11] and [15]), with the exception of relativistic corrections to the former, which become important for near-critical fields. In this section we derive the form for photon angle redistribution to be used in the code based on the on-resonance cyclotron emission approximation.

Let us represent the photon's shift in energy and angle on scattering through the following schematic:

$$[(k^0, \mu), (p_e, n = 0)]_i \rightarrow [(k^0, \mu), (p_e, n)]_f,$$

where all quantities are evaluated in the lab frame, the photon's 4-momentum is k , $\mu \equiv \hat{k} \cdot \hat{B}$, and the electron energy states are characterized by (p_e, n) .

Use will be made of the exact expression for cyclotron emission rate in a frame in which $p_z = 0$ (note, however, that this is *not* the “rest frame”) and Lorentz transform the solid angle of emission to a frame in which the “intermediate” electron has $p_z = k'^0 \cos \theta$, which is the boost imparted on the intermediate state by the requirement of parallel momentum conservation in a frame in which the electron is *originally* at rest. Let us label the quantities in the former frame with “'” and in the latter with “.” Then the re-emission process:

$$(p'_e = q, n)_{\text{int}} \rightarrow [(k'^0, \mu'), (p'_e, n)]_f,$$

has a corresponding schematic,

$$(p''_e = 0, n)_{\text{int}} \rightarrow [(k''^0, \mu'), (p''_e, n)]_f,$$

in the “0p-frame” of the intermediate electron.

In the 0p-frame, the differential transition rate for cyclotron emission from an excited Landau level n to the ground state, averaged over initial and summed over final spin states, is given by equation (15), where n will correspond to the intermediate level n_{int} . This equation is consistent with our framework, since we do not consider spin-dependent angular redistribution that has been suppressed in equation (15) by making the corresponding sum over final electron spin states.

Strictly speaking, equation (15) is valid only for transitions $n_{\text{int}} \rightarrow 0$. However, we will make use of it for transitions $n_{\text{int}} \rightarrow n_f (< n_{\text{int}})$ by making the substitution $n = n_{\text{int}} - n_f$ above. The motivation for this is that for low-lying states ($n_{\text{int}} < 4$) in low fields $B/B_{\text{crit}} \ll 1$, numerical evaluation of the relativistic transition rates shows that these rates possess an angular distribution (averaged over spin and polarization) that closely resembles that of the ground state.

Now, to evaluate the angular distribution for re-emission in the primed reference frame, one must substitute $\mu'_f(\mu'_f)$ in equation (15) with its explicit dependence as given by the Lorentz transform of a particle on the light cone, $\mu'' = (\mu' - \beta)/(1 - \beta\mu')$, divide $(d\Gamma_{n \rightarrow 0})/d\mu'_f$ by γ to account for time dilation and multiply it by the Jacobian:

$$\left| \frac{\partial \mu''}{\partial \mu'} \right| = \left[\frac{1}{\gamma(1 - \beta\mu')} \right]^2 \quad (55)$$

to obtain $(d\Gamma_{n \rightarrow 0})/d\mu'_f$.

The Lorentz transformation is done from a frame in which $p'_z = k'^0 \cos \theta$ to a frame in which $p'_z = 0$. Thus $E'' = \gamma(E' - \beta p'_z)$, $0 = \gamma(p'_z - \beta E')$, and therefore

$$\gamma = \left(\frac{1 + 2nB' + p'^2_z}{1 + 2nB'} \right)^{1/2} \quad \text{and} \quad \beta = \frac{p'_z}{(1 + 2nB' + p'^2_z)^{1/2}}. \quad (56)$$

Further, by noting that

$$E'' = \frac{E' - p'_z \mu}{\gamma(1 - \beta p'_z)}, \quad \text{and} \quad v'^2 \rightarrow \frac{v'^2}{[\gamma(1 - \beta \mu')]^2},$$

the desired form is obtained (compare with eq. [15]):

$$\begin{aligned} \frac{d\Gamma_{n \rightarrow 0}}{d\mu}(\mu) &= \frac{\alpha}{(n-1)!} \frac{1}{(E - p_z \mu)^2} \frac{2nB'}{E} e^{-Z} Z^{n-1} \\ &\times \frac{1}{(1 + \xi)^2} \left(1 + 2nB' - \frac{1}{\xi} \right), \end{aligned} \quad (57)$$

where

$$\xi \equiv \frac{1 - 2nB'v^2}{(E - p_z \mu)^2},$$

and all the quantities are understood to be evaluated in the primed frame (i.e., in the frame in which the electron is *initially* at rest). The grouping in equation (57) is done to exhibit the transformation properties of the rate. Note that

the quantity

$$\frac{v^2}{\{E - p_z \mu\}^2} = \frac{v'^2}{\{E|_{p=0,n} \gamma(1 - \mu' \beta)\}^2} = \frac{v'^2}{E'^2}$$

is a Lorentz invariant for transformations along the z direction. This is a reflection of the fact that as long as the electron is in an excited Landau level in a near-critical field, it remains relativistic even if $p_z = 0$, and therefore a “rest frame” doesn’t exist for such electron. The first nonconstant factor in equation (57) accounts for the transformation properties of the solid angle, the second factor absorbs the time dilation, and the rest of the expression explicitly depends on the invariant, ξ , mentioned above. Equation (57) is easily evaluated numerically and is good for fields comparable with B_{crit} .

For lower field strengths equation (15) may be simplified. Consider the classical limit

$$Z \simeq \frac{1}{2} n^2 B' v^2 \ll 1, \quad k^0 \simeq nB' \ll 1, \quad \text{and} \quad E_n \simeq 1 + nB',$$

to obtain

$$\frac{d\Gamma_{n \rightarrow 0}}{d\mu} \approx \frac{\alpha}{4(n-1)!} \frac{2nB'}{(1 + 2nB')^{3/2}} \left(\frac{n^2}{2} B' \right)^{n-1} (1 + \mu^2) v^{2(n-1)} \quad (58)$$

in this limit through a procedure similar to the one used above. The simplified result:

$$\begin{aligned} \frac{d\Gamma_{n \rightarrow 0}}{d\mu} &\propto \left[\frac{1}{\gamma(1 - \beta\mu)} \right]^2 \times \left[1 + \left(\frac{\beta - \mu'}{1 - \beta\mu'} \right)^2 \right] \\ &\times \left[\frac{v'}{\gamma(1 - \beta\mu')^2} \right]^{2(n-1)}, \end{aligned} \quad (59)$$

is readily derived. Note that the first factor on the right corresponds to the Jacobian, the second one may be interpreted as the redistribution for first harmonic photons, and the last one is a correction for higher harmonic photons.

As expected, both equations (59) and (57) give preferential forward scattering. The value of k'^0 is taken to be the relativistic resonant *absorption* energy:

$$k'^0_{\text{res}} = \frac{(1 + 2nB' \sin^2 \theta')^{1/2} - 1}{\sin^2 \theta'} \quad (60)$$

in equation (57) above, or its classical counterpart $k'^0_{\text{res}} \simeq nB'$ in equation (59). Finally, the angle of the scattered photon determines its final energy through the kinematic relation given in § 5.4.

6. CYCLOTRON LINE FEATURES IN HARD CONTINUA

We will now utilize the scheme constructed in the preceding sections to investigate the formation of cyclotron lines in hard X-ray continua and assess the effect of geometry and field strength on the ensuing line features. The use of a hard continuum (see eq. [61]) was originally intended to model galactic GRB scenarios. However, the results contained within are also permissible for X-ray pulsars having

a hard continuum at the line (i.e., low magnetic field or high turnover energy).

Our aim in producing cyclotron line spectra in such continua is twofold. First, a fairly flat X-ray continuum shone on magnetized plasmas allows for an easier interpretation of the effects of photon spawning, relativistic angular redistribution, and geometry on the formation of the fundamental cyclotron line feature. The theoretical spectra thus generated greatly aid in the interpretation of cyclotron lines formed in softer continua such as those pertaining to X-ray pulsars. Second, motivated by the discovery of double harmonic line features at ≈ 20 and 40 keV in GB 880205 and the research that this event spurred (Wang et al. 1989; Alexander & Mészáros 1989), we have sought to study models in the parameter space evidenced by this object with a fully relativistic formulation, including the effect of plasma geometry.

Regarding GB 880205, Wang et al. (1989) note that the two lines are relatively narrow and have similar line strengths. The first remark indicates a low electron temperature relative to the continuum hardness that Lamb et al. (1990) find to be physically motivated (§ 3.2). Additionally, the process of photon spawning is invoked by the former group to “fill in” the first harmonic so that at $\mu \approx .31$ the first two harmonics have approximately equal line profiles. The inferred value of the column density, $N_e = (1.2 \pm .6) \times 10^{21} \text{ cm}^{-2}$, corresponds to a continuum optical depth of $\tau_c = (0.6 - 1.2) \times 10^{-3}$.

6.1. Model Parameter Selection

The incident photon number spectra depend on energy through a simple power-law-type function:

$$\frac{dN_\omega}{d\omega d\mu} \propto \omega^\alpha \quad \text{with} \quad \alpha = -1. \quad (61)$$

Photon injection is isotropic, although the emergent spectra are not. This fact suggests that to obtain self-consistent results one should inject the photons in a non-isotropic fashion. The results of such an exercise will be presented in a separate paper along with a detailed study of the angular distribution of the emergent radiation and the results of beamed angular injection perpendicular to and along the $\mathbf{B}$ field.

Two geometries for the scattering region are considered: plane parallel slab (with the magnetic field parallel to the slab normal) and cylindrical (with the field parallel to the cylinder axis). The continuum photon spectrum is incident from a source at the slab midplane or along the cylinder axis. The escaping photons are accumulated in four ranges of $\mu \equiv \cos \theta$.

Three parameters characterize the line: the electron temperature $T_e = \frac{1}{4}B/B_{\text{crit}}$ (physically motivated temperature Lamb et al. 1990), the magnetic field B (assumed uniform), and the minimum value of continuum optical depth τ_c (in the direction parallel to $\mathbf{B}$ for a slab and perpendicular to $\mathbf{B}$ for a cylinder). With the exception of the optical depth progression figures, Figures 4 and 5 below, the continuum optical depth in all other spectra is $\tau_c = 10^{-3}$. Moreover, the magnetic field progression plots have $B' \equiv B/B_{\text{crit}} = .04, 0.1, \text{ and } 0.24$ (1.7, 4.4, and 11. TG, respectively).

Each spectrum results from 50,000 isotropically injected photons redistributed into four angle bins. The spectra are

thus constrained by limited statistics. Photon depletion at line center is common and unavoidable with the limited samples. Crosses indicate parts of the spectrum where photon depletion has occurred. Given that the energy grid has N_E sampling energy points and N_μ angle bins, photon depletion anywhere inside the line puts a rough lower limit on the optical depth at that energy (assuming *unbiased* injection):

$$\tau > \tau_{\min} = -\ln\left(\frac{N_\mu N_E}{N_{\text{inj}}}\right). \quad (62)$$

For $N_E = 80$, $N_\mu = 4$ and $N_{\text{inj}} = 5 \times 10^4$, this expression gives $\tau_{\min} \approx 5$.

6.2. The Scattering Profile

First, some simple remarks about the profile function are in order. The shape of the profile results from an interplay between relativistic broadening and the dependence of the cross section on the photon's *rest frame* energy shift from the resonant energy (§ 5.3). The latter depends on momentum through the Lorentz boosted value of μ : $\omega^{\text{res}}[\mu'(p)]$ (eq. [37]).¹¹ Additionally, note that the fractional number of electrons close to $p = 0$,

$$dN_e = |f_e(p)|_{p \sim 0} dp,$$

is larger at lower temperatures so that more electrons are able to see a nearly resonant photon, k , for $T_e \ll 1$. As the equilibrium temperature of the electrons in the scattering layer scales directly with the cyclotron frequency, equation (4), and since the (absorption) cross section goes as E^{-1} (see eq. [11]), one should expect the profiles at low (i.e., well under critical) fields to be strongly dependent on the detailed behavior of the cross section near resonance.

Figures 1 and 2 provide a valuable tool to interpret and understand all the modeled line features. First, note that the value of the profiles near resonance is in fact stronger at lower fields and the one-dimensional Doppler-broadened profiles are widest at small angles to the field, where the photons see the largest spread in electron momenta. In addition, since the nominal estimate of the electron temperature scales with the field strength, higher field models should have not only a broader profile but also a good measure of relativistic Lorentz beaming along the field (§ 5.6). Figure 1, for instance, has a mildly relativistic temperature $T_e \approx 31$ keV, while Figure 2 has a nonrelativistic temperature $T_e \approx 5$ keV.

A second, perhaps unexpected, observation can be made of the profiles (see Fig. 3) for large angles to the $\mathbf{B}$ field (i.e., $\mu < .25$): these functions need not peak at resonance. Instead, the profile peaks at the relativistic energy cutoff. This has a simple but interesting explanation. Following the analysis of the profile's integrand behavior in § 5.3.2, recall the form of the integrand:

$$\mathcal{A}rg(p, k^0, \mu) dp \equiv \frac{1}{\mathcal{D}} f_e(p)(1 - \mu\beta)\sigma_{rf}(k'^0, \mu') dp,$$

and note that the largest differential contribution to the

¹¹ Recall our convention from § 5: primes denote rest frame quantities, unprimed symbols the quantities in the lab-frame.

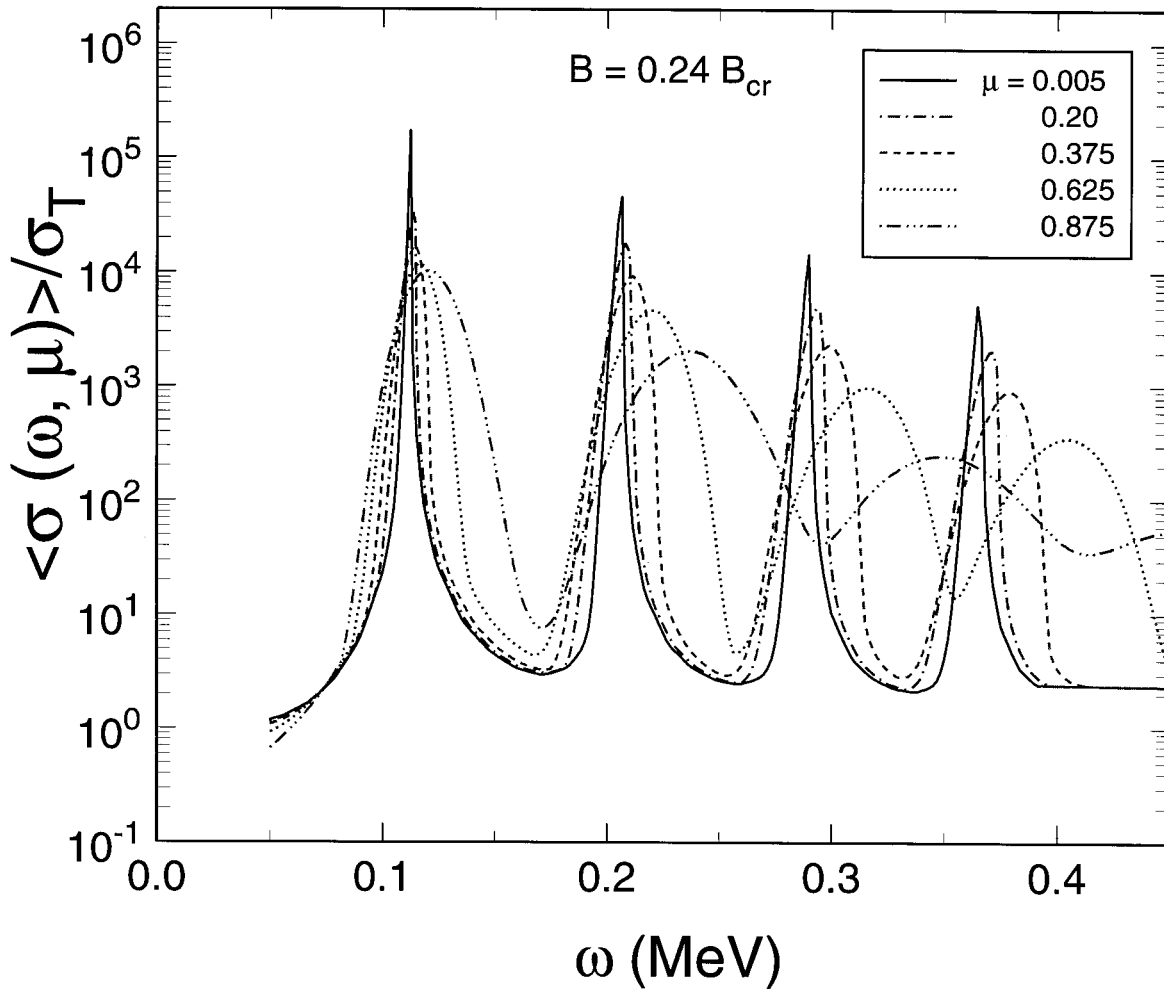


FIG. 1.—Total profile function for $B' = 0.2436$ (10.75 TG). $\mu \equiv \cos \theta$. The sharpest profile corresponds to $\mu \sim 0$. Note the very sharp “down step” at the relativistic cut off energy for $\mu = 0.2$.

profile comes from where $\mathcal{A}rg$ attains its phase-space maximum (i.e., for the largest possible range of p values). For $\mu < \sim .2$ this condition is met at the relativistic energy cutoff and not at resonance. Moreover, since the profiles are stronger at lower fields, one expects the sharply peaked nature of the profiles near the relativistic energy cutoff to be more sensitive at lower fields than at larger ones. This fact is evidenced in the profile graphs. Both the resonant energy and the relativistic cutoff energy are chosen as sampling points in each of the profile figures.

6.3. Geometry and Optical Depth Progression

We are now ready to discuss the optical depth progression plots (Figs. 4 and 5) and the variation in the line shapes with angle and geometry.

6.3.1. Second and Higher Harmonic Features

The dynamics of line formation are most complex at the fundamental energy, and we will temporarily defer its discussion. Higher harmonic line features, on the other hand, may be understood as “almost” absorption features, since at low fields the electrons prefer to de-excite in single perpendicular momentum quanta. Consequently, second and higher harmonic features closely follow the shape dictated

by the profile function. Figures 1 and 2 show all the relevant harmonic energies for the formation of the cyclotron spectra displayed here.

From left to right and from top to bottom, Figures 4 and 5 show the optical depth progression of line features for a magnetic field $B = 1.7$ TG in slab and cylindrical geometries, respectively. Already at $\tau_c = 1.0 \times 10^{-4}$ remarkable differences become apparent. Second and higher harmonic features at large angles (*bottom graphs*) are stronger for slab geometries than their cylindrical siblings. This is an expected feature: for photons emerging from a slab, the optical depth is largest at large angles

$$\tau_c^{\text{sl}}(\theta) = \tau_c / \cos \theta ,$$

while for photons in a cylindrical scattering region the optical depth is largest at small angles

$$\tau_c^{\text{cy}}(\theta) = \tau_c / \sin \theta .$$

Moreover, since the profiles are broadest at small angles and deepest at large angles, the slabs show the deepest features for second and higher harmonics at low depths, $\tau_c \leq 3 \times 10^{-4}$, in all but the very small angle bin. Cylinders, on the other hand, present slightly wider features

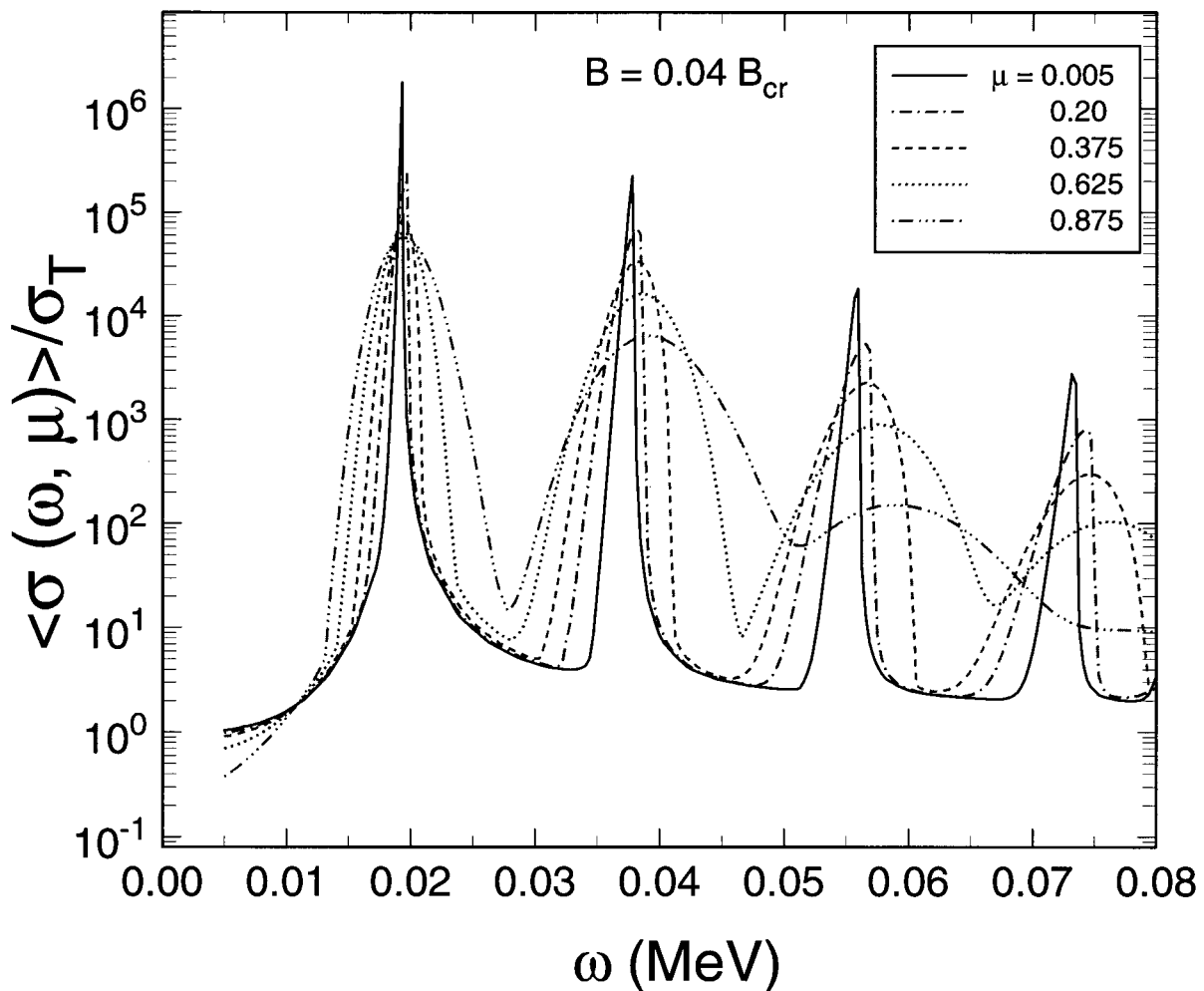


FIG. 2.—Total profile function for $B' = 0.04$ (1.7 TG). The tallest, narrowest profile corresponds to $\mu = 0.005$. The curves shown correspond to the same $\mu (\equiv \cos \theta)$ parameter values as those of Fig. 1.

at small angles and larger depths because the profiles are broad but weaker there. This absorption-like trend becomes more pronounced as the optical depth gets larger.

6.3.2. The Fundamental Feature and the Consequences of Photon Spawning

By contrast, the features that result from first harmonic scattering have no straightforward interpretation. The effect of multiple resonant scattering on the line features is felt mostly at the fundamental harmonic energy. Higher harmonic photons are more likely to undergo single resonant Raman-Landau scattering, leaving the electron in an excited Landau level, effectively destroying the photon. Table 1 supports this assessment by considering the field strength-dependent branching ratios for several photon spawning processes. The first two ratios conserve the resonant photon and do not produce spawned photons. The second two ratios spawn first harmonic photons, and the last process spawns a second harmonic photon from the scattering of a third harmonic.

Subsequent decay of excited states produces spawned photons that effectively amount to additional photon injection *shortward* (i.e., anharmonic spacing) of the fundamental

cyclotron energy for lower fields. The contribution to the emergent spectra from spawned photons may be significant indeed. Table 2 illustrates this fact by considering the number of spawned first harmonic photons relative to the total number of emergent photons, as a function of optical depth and geometry. Note that the relative numbers of spawned photons are slightly lower for a cylindrical geometry.

Escaping photons in slab scattering regions tend to redis-

TABLE 1
PHOTON SPAWNING BRANCHING RATIOS^a

Branching Ratio $\mu = 1/\sqrt{2}$	n	$B' = .04$	$B' = .24$
$\mathcal{S}_0/\sigma_2$	2	0.074	0.238
$\mathcal{S}_0/\sigma_3$	3	0.013	0.118
$\mathcal{S}_1/\sigma_2$	2	0.926	0.762
$\mathcal{S}_2/\sigma_3$	3	0.859	0.608
$\mathcal{S}_1/\sigma_3$	3	0.128	0.273

^a In the notation of § 6.3.1: σ_n denotes the *total* cross section summed over all levels l for a photon having $k^0 = \omega_n^{\text{res}}(\pi/4)$; and $\mathcal{S}_l$ denotes the cross section into final Landau level l . See text for discussion.

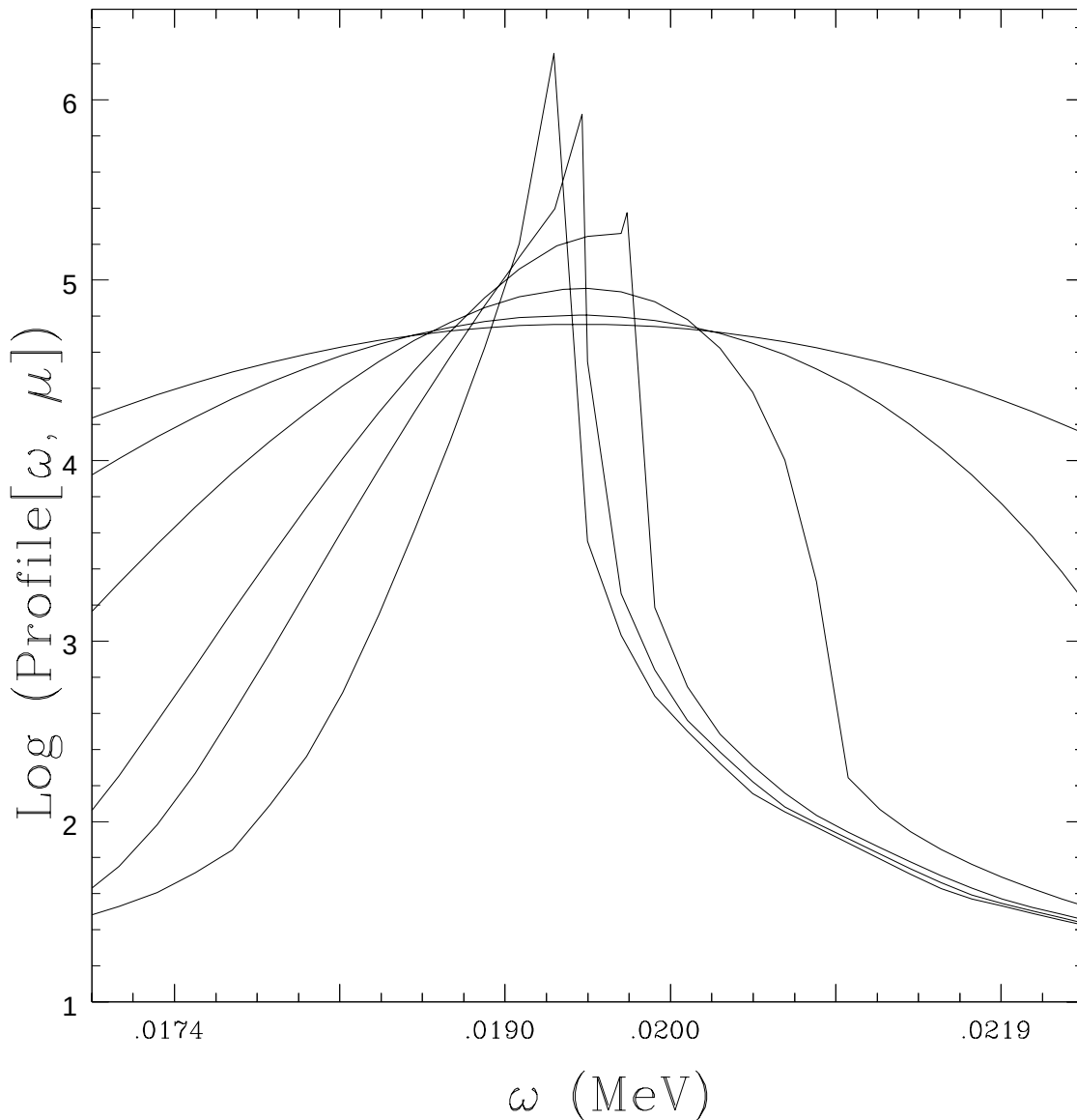


FIG. 3.—Close-up of the total profile function for $B' = 0.04$. Note that at this field, a cusp occurs instead of the down step at the relativistic cutoff energy for $\mu = 0.2$ (see text for explanation).

tribute to the line wings from all angles and produce large broad emission line wings at small angles that are enhanced by the extra injection from spawned photons. This mechanism also manifests itself in cylinders, although less effectively because of the higher optical depth at those angles.

TABLE 2

RELATIVE NUMBER OF SPAWNED FUNDAMENTAL PHOTONS^a

$\tau_c [\times 10^{-3}]$	0.1	0.3	1.0	3.0
$N_{sp}^{sl}/\{N_{inj} + N_{sp}\} \dots\dots$	10%	16%	23%	32%
$N_{sp}^{cy}/\{N_{inj} + N_{sp}\} \dots\dots$	7%	12%	19%	28%

^a As a function of optical depth and geometry for a field $B' = 0.04$. The superscript refers to slab and cylindrical geometries.

The broad line wings that emerged in our study are a new feature of theoretical cyclotron line models.

In addition, even if the injection is isotropic, a strongly anisotropic radiation field ensues from relativistic photon redistribution and from the large numbers of spawned photons that a *hard* continuum produces.

Moreover, by noticing that the profiles are narrowest and deepest at large angles and cylinders have the most favorable escaping conditions at those angles, the formation of the *emission* line for $\mu < 0.25$ in Figure 5 is explained. Filling in by angle redistribution from small angles to large ones is much less effective in a slab geometry, and no single emission feature is formed.

Note that for $\tau_c = 1 \times 10^{-3}$ and for angle bin $\mu > 0.75$, cylinders produce first and second harmonic features with very similar equivalent widths to those required to fit the GB 880205 observation. Therefore, the consideration of geometry suggests that the model produced by Wang et al.

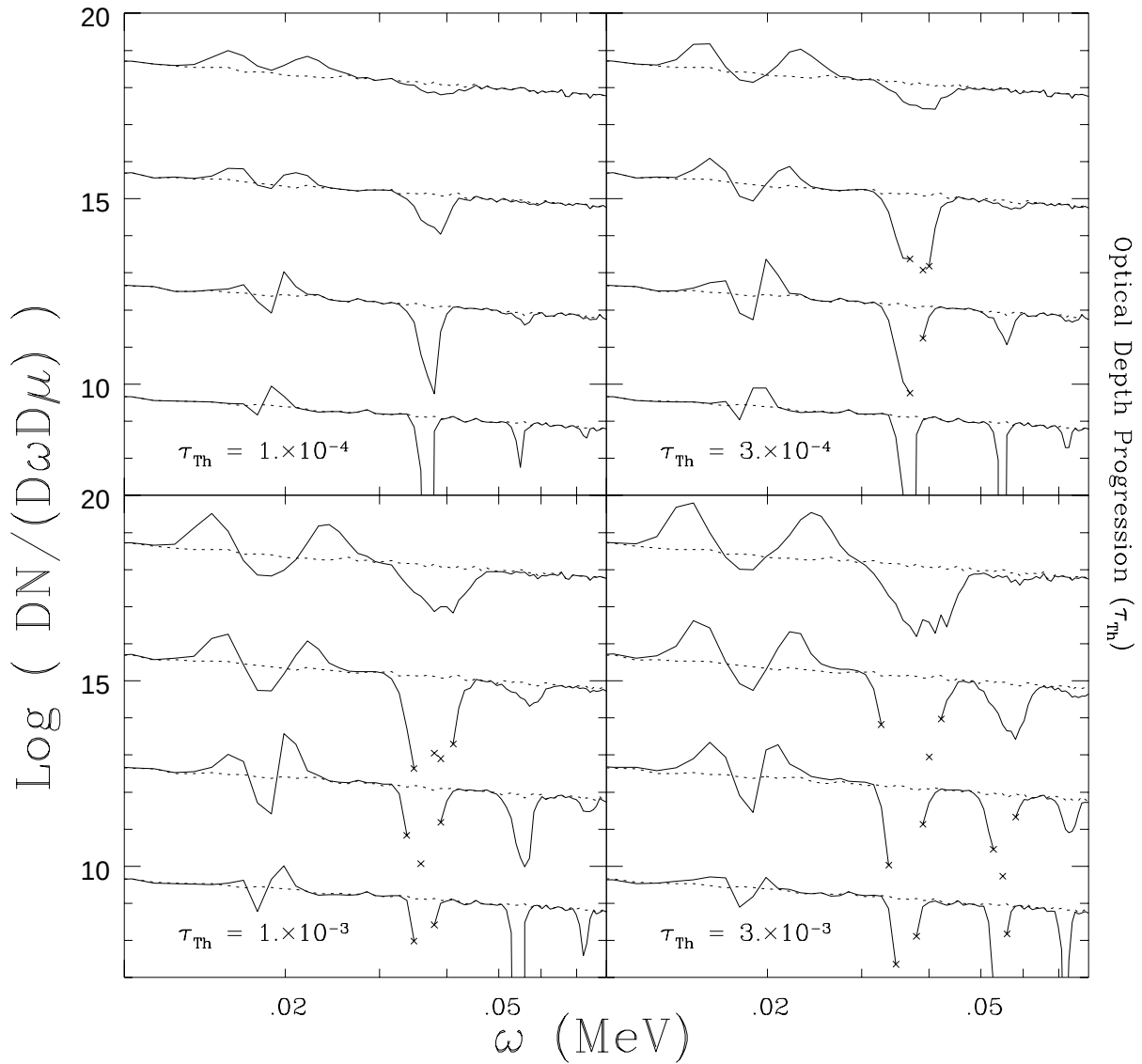


FIG. 4.—Slab geometry, optical depth progression of cyclotron line features. Each plot shows angle-dependent model photon spectra. On each quadrant the bottom curve is the spectrum emerging from $\mu < 0.25$, followed by spectra from $0.25 < \mu < 0.50$, from $0.5 < \mu < 0.75$ and from $\mu > 0.75$ for the top curve. The magnetic field strength is $B' = 0.04$ (1.7 TG). The run results from 50,000 photons injected isotropically. Dotted line: injected continuum, power law with $\alpha = -1$. Solid line: output scattered spectrum. Crosses: areas where photon depletion has occurred; the model points are indicated as crosses for ease of reading. τ_c is indicated in each quadrant, $T_e = \frac{1}{4}\omega_{\text{cyc}}$. The flux normalization is arbitrary. The spectra emitted near the neutron star are local spectra emitted near the neutron star surface and do not include general relativistic effects such as red shift or light bending.

(1989) is not unique in its production of similar strength first and second harmonic line features.

6.4. Geometry and Field Strength Progression

The dependence of cyclotron line spectra on magnetic field strength is illustrated in Figure 6. For fields approaching B_{crit} , higher harmonic resonant scattering (i.e., leaving the electron back in the ground level) becomes important, and low harmonic line shapes could be expected to imitate the fundamental harmonic line features. Additionally, Table 1 makes it apparent that at those fields a second harmonic photon is more likely to survive a scattering event and that third harmonic photons tend to spawn second harmonic photons, thus imitating the behavior of spawned fundamental photons at lower fields. The third and higher harmonic line features will qualitatively follow the “absorption-like”

trends detailed in § 6.3.1, compounded by a mild relativistic beaming along the $\mathbf{B}$ field.

The formation of the fundamental line, however, is strongly dictated by angular redistribution in addition to photon spawning, and the spawned second and higher harmonic photons differ in their angular redistribution from the corresponding distribution for spawned fundamental photons. In order to make a qualitative estimate of the latter statement, recall the discussion that followed equation (59) in the previous section. Ignoring relativistic beaming, the angular redistribution of higher harmonic photons is $\propto \sin^{2(n-1)} \theta$. This constitutes a fundamental difference on the kinematics of line formation for higher harmonic lines at high fields. Thus although one may have expected to find some similarity between the fundamental feature and the second harmonic line at large fields, this is

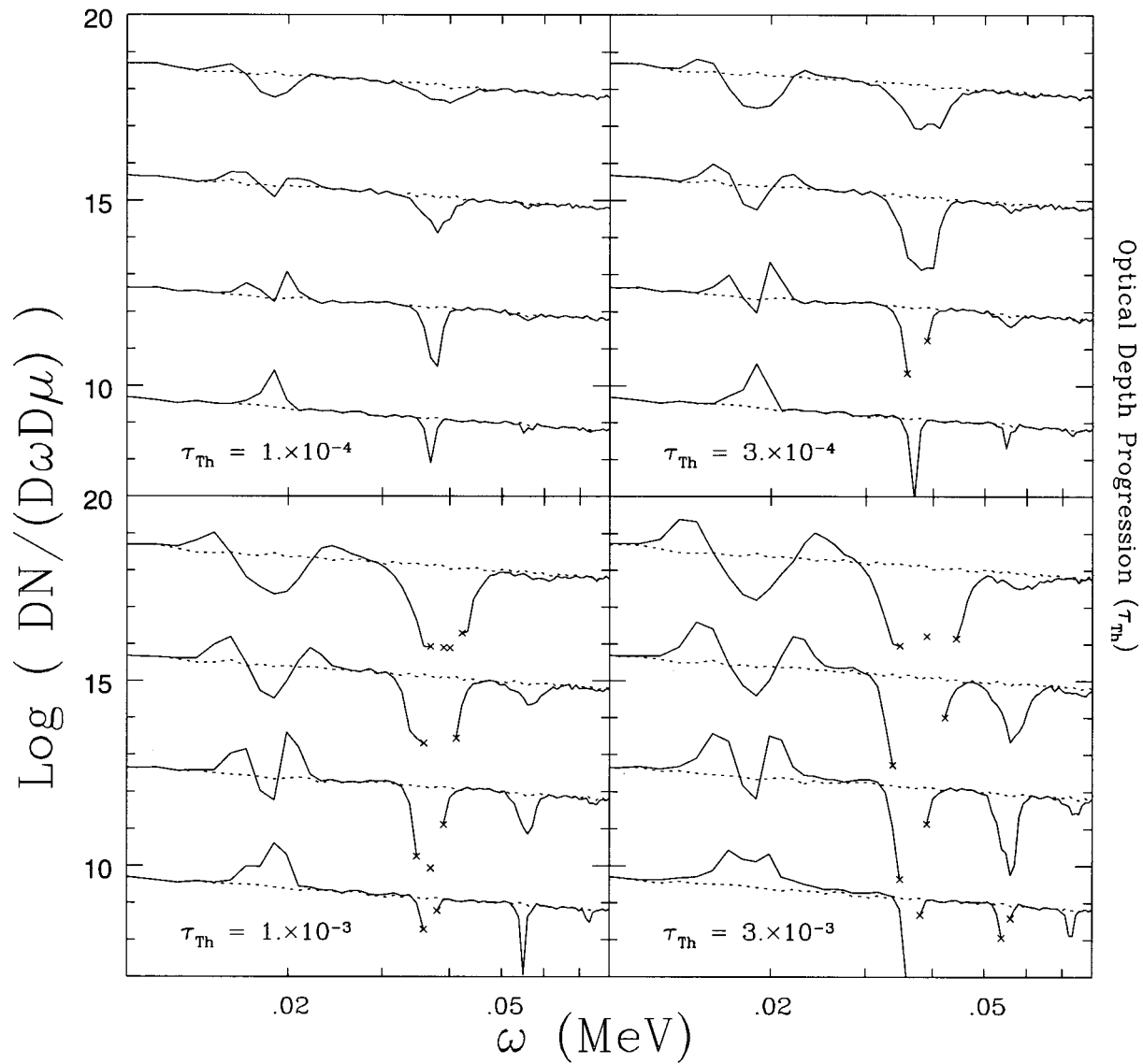


FIG. 5.—Cylinder geometry, optical depth progression of cyclotron line features. The convention follows that from the slab progression plot. The magnetic field strength is $B' = 0.04$ (1.7 TG). The run results from 50,000 photons injected isotropically. Dotted line: injected continuum, power law with $\alpha = -1$. Solid line: output scattered spectrum. Crosses: areas where photon depletion has occurred; the model points are indicated as crosses for ease of reading. τ_c is indicated in each quadrant, $T_e = \frac{1}{4}\omega_{cyc}$. The flux normalization is arbitrary.

not borne out from our numerical results for $B' = 0.24$ or from the analysis above.

As expected, all spectra from intermediate angles $0.75 > \mu > 0.25$ are somewhat similar owing to their roughly equivalent optical depths. Thus the determination of scattering region geometry from shape of the cyclotron line features is ambiguous at these angles. The strongest dependence of the lines on geometry occurs at large $\mu < 0.25$ and small $\mu > 0.75$ values of the emergent angle to the field where, for example, absorption features in one geometry become emission features in the other.

In general, at larger optical depths the second harmonic lines become weaker, as expected from the profile behavior and from the spawned photon contribution. These features also become asymmetric at large angles as motivated by the asymmetric profile shape near the relativistic cutoff energy. The slab models yield the broadest lines at those angles.

At $B' = 0.24$, the fundamental features seen at small

angles to the field become so broad and weak that their detection may at present be impossible. Already for intermediate field strengths, slabs *emit* energy at those angles as the absorption dip at the fundamental energy is overwhelmed by very broad emission “line wings.”

The fundamental line features at larger angles ($\mu < 0.5$) are strongly asymmetric and, a single emission line wing now appears in the blue side of the fundamental. This feature is more pronounced for cylinders where the optical depth at those angles is smaller. Angular redistribution from small angles to large ones may account for such behavior but this issue has not been resolved at present. Moreover, there is no reason to believe that at these large fields the nominal estimate of electron temperature (Lamb et al. 1990) will give the correct result. Further analysis, including the angular dependence of the emergent radiation field and detailed balance of the electron energetics, will help to elucidate these and other issues.

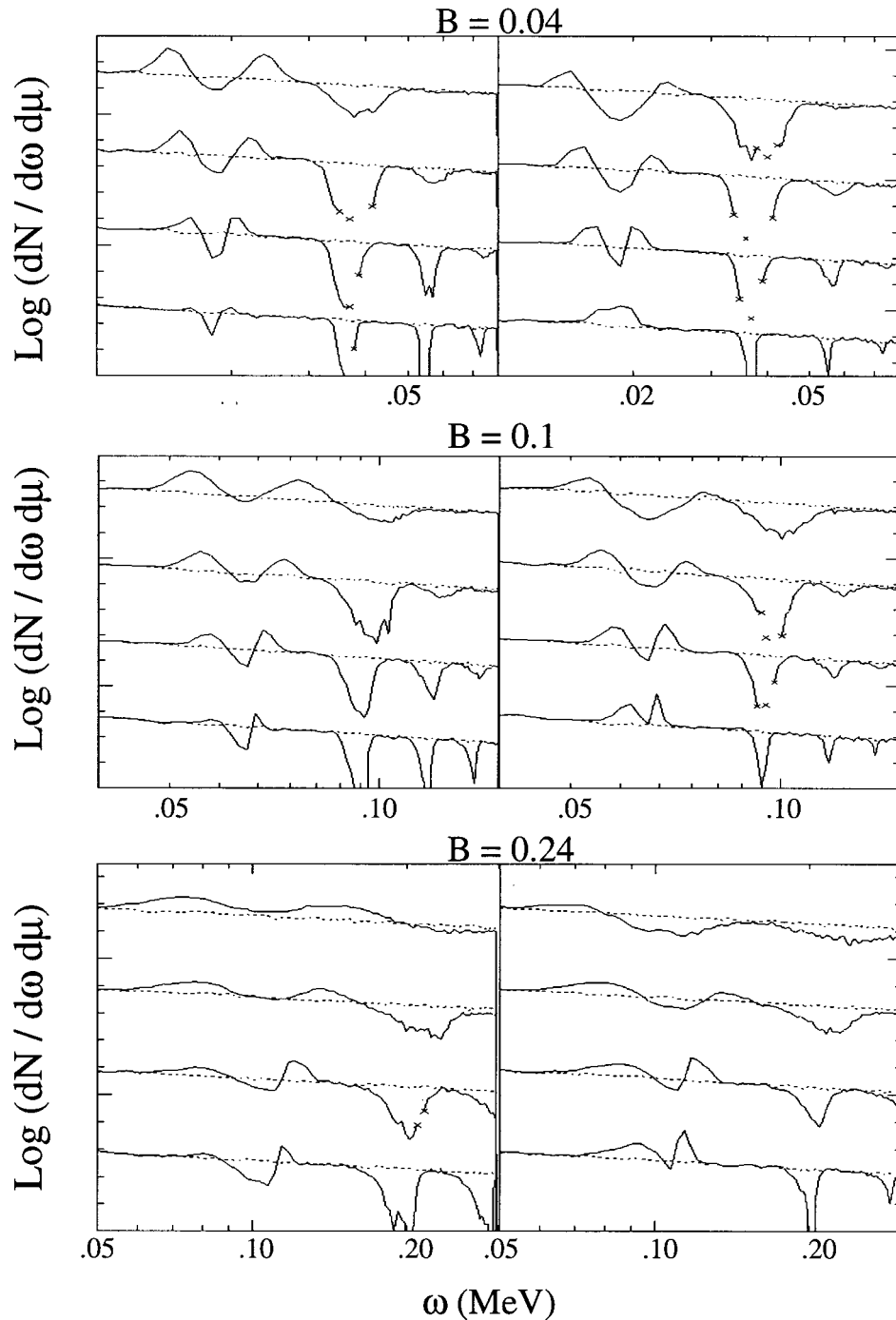


FIG. 6.—Magnetic field progression of cyclotron line features in slab and cylinder geometry. Each plot shows angle dependent model photon spectra: the bottom curves are the spectrum emerging from $\mu < 0.25$, followed by spectra from $0.25 < \mu < 0.50$, from $0.5 < \mu < 0.75$ and from $\mu > 0.75$ for the top curve. Dotted line: injected continuum, power law with $\alpha = -1$. Solid line: output scattered spectrum. Crosses: areas where photon depletion has occurred; the model points are indicated as crosses for ease of reading. Spectra for slab geometry are on the left and for cylinder geometry on the right. For all spectra, $\tau_{\text{th}} = 3 \times 10^{-3}$.

7. SOME FINAL REMARKS

The model laid out in this article treats the formation of cyclotron lines in the low-density/high-field limit (§ 3.3) for internally irradiated plasmas. Its primary features, use of relativistic angle redistribution of photons, use of the magnetic QED Compton-scattering cross section with inclusion of natural line widths, treatment of up to four harmonic photons, and allowance of spatial photon diffusion in arbi-

trary geometries, make this program a solid tool to model the intended regime.

The results presented in the preceding section demonstrate the usefulness of the program, although the low field strength, $B = 1.7$ TG, provides a marginal case of the intended regimen. In practice, if one restricts oneself to low optical depths ($\tau_c < 0.01$), the model is on solid ground. Our approximation for angular redistribution (i.e., the on-shell

approximation) also is only valid for spectra that are optically thin in the line wings, otherwise the off-shell component may be important (J. C. Wang 1996, private communication).

Finally, we must point out that to obtain more physical results, self-consistent models of nonisotropic radiation fields must be carried out. A first exercise toward that goal should be the study of injection of beamed radiation perpendicular and parallel to the field. This task will be investigated in a future paper. Lastly, we would like to point out that models of cyclotron lines in hard continua are applic-

able to X-ray pulsars having low magnetic fields or high turnover energies where the cyclotron harmonic features lie in the hard spectral region. These models also greatly aid the interpretation of the harmonic line features in softer continua such as those belonging to typical X-ray pulsars.

We would like to thank R. Sina for supplying the code to calculate the cross section using Sokolov-Ternov electron wave functions and A. Szalay for allowing us access to the supercomputer resources at Johns Hopkins University.

APPENDIX

A CALCULATION OF THE SCATTERING CROSS SECTION

The two terms below represent Compton scattering from Feynman diagrams in which the initial (final) photon is first absorbed (emitted) and then the final (initial) photon is emitted (absorbed). The matrix elements are related by the crossing symmetry substitutions,

$$S^{(1)} \rightleftharpoons S^{(2)}, \quad \mathbf{k}_i \rightleftharpoons -\mathbf{k}_f, \quad k_i^0 \rightleftharpoons k_f^0 \quad \text{and} \quad \epsilon_i \rightleftharpoons \epsilon_f. \quad (\text{A1})$$

For each diagram there is an S -matrix element:

$$S_{fi}^{(1)} = -e^2 \int d^4x d^4x' [\bar{\Psi}_f(x') \gamma_\mu A_f^\mu(x') G(x', x) \gamma_\nu A_i^\nu(x) \Psi_i(x)], \quad (\text{A2})$$

$$S_{fi}^{(2)} = -e^2 \int d^4x d^4x' [\bar{\Psi}_f(x') \gamma_\mu A_i^\mu(x') G(x', x) \gamma_\nu A_f^\nu(x) \Psi_i(x)], \quad (\text{A3})$$

with

$$p_z^n = p_i \pm k_{i,f}^0 \cos \theta_{i,f} \quad \text{and} \quad E_n^2 = m^2 + p_z^n + 2neB, \quad (\text{A4})$$

the *respective* intermediate electron momenta and energy, and where the intermediate states are also characterized by the Landau levels n and spins s_n .

For the evaluation of the S -matrix elements, use of the Landau gauge,

$$\mathbf{A}(x) = (0, Bx, 0) \quad \text{and} \quad A^0(x) = 0, \quad (\text{A5})$$

along with Cartesian coordinates, proves to be the simplest calculational choice. In this *Cartesian-Landau gauge*, electron wave functions are given by:

$$\begin{aligned} \Psi(x) &= \begin{pmatrix} S_1 & I_{n-1}(\eta) \\ S_2 & I_n(\eta) \\ S_3 & I_{n-1}(\eta) \\ S_4 & I_n(\eta) \end{pmatrix} e^{i\xi(Et - p_y y - p_z z)}, \\ &\equiv \phi_{\xi,q} e^{i\xi Et}, \end{aligned} \quad (\text{A6})$$

with

$$2neB = p_n^2 \equiv E^2 - m^2 - p_z^2. \quad (\text{A7})$$

Here $\eta \equiv (eB)^{1/2}[x + \xi p_y/(eB)]$, and ξ is the sign of the energy characterizing “electron” $\xi = +1$ and “positron” $\xi = -1$ states in the nonrelativistic limit. E and p have their usual interpretation for electron states and minus the corresponding operator eigenvalue for positron states.

The I_n ’s are related to the Hermite polynomials, H_n , through

$$I_n(\eta) = \left(\frac{e^{-\eta^2}}{\sqrt{\pi} 2^n n!} \right)^{1/2} H_n(\eta). \quad (\text{A8})$$

Furthermore, the normalization constants, S_i ’s, must satisfy the following relation (Melrose & Parle 1984):

$$(S_1, S_2, S_3, S_4) \mathcal{T} = 0, \quad (\text{A9})$$

with

$$\mathcal{F} \equiv \begin{pmatrix} -\xi E + m & 0 & \xi p_z & -ip_n \\ 0 & -\xi E + m & ip_n & -\xi p_z \\ \xi p_z & -ip_n & -\xi E - m & 0 \\ ip_n & -\xi p_z & 0 & -\xi E - m \end{pmatrix},$$

independently of the choice of coordinate or gauge.

States of the Sokolov-Ternov (1968) spin operator, equation (10), also commute with the radiative corrections to the Dirac-Landau equation, and this enables the introduction of natural line widths to the scattering cross section if simultaneous eigenstates of μ are chosen for the electron wave functions. Sokolov and Ternov's choice for the spin constants is

$$\begin{pmatrix} S_1 \\ S_2 \\ S_3 \\ S_4 \end{pmatrix} = \frac{1}{2^{(3/2)}} \begin{pmatrix} \sqrt{1 + (\tau_z m/E_\perp)} [\sqrt{1 + (p_z/E)} + \xi \tau_z \sqrt{1 - (p_z/E)}] \\ \tau_z \sqrt{1 - (\tau_z m/E_\perp)} [\xi \tau_z \sqrt{1 - (p_z/E)} - \sqrt{1 + (p_z/E)}] \\ \sqrt{1 + (\tau_z m/E_\perp)} [\sqrt{1 + (p_z/E)} - \xi \tau_z \sqrt{1 - (p_z/E)}] \\ \tau_z \sqrt{1 - (\tau_z m/E_\perp)} [\sqrt{1 + (p_z/E)} + \xi \tau_z \sqrt{1 - (p_z/E)}] \end{pmatrix}, \quad (\text{A10})$$

where $\tau_z = \pm 1 = 2s_z$ refers to spin parallel (antiparallel) to the field and $E_\perp = (m^2 + 2neB)^{1/2}$.

With the following definition for the "generalized current", $[J(k)]^\nu$:

$$[J_{qr}^{(\xi\eta)}(k)]^\nu \epsilon_\nu^j = \int d^3x e^{\pm i\mathbf{k} \cdot \mathbf{x}} \bar{\phi}_{\xi,q}^{s_q}(x) \gamma^\nu \epsilon_\nu^j \phi_{\eta,r}^{s_r}(x), \quad (\text{A11})$$

where the $\phi_{\xi,q}$'s are the time-independent part of the bispinor; the total S matrix amplitude is:

$$\begin{aligned} S_{fi} = & -i\alpha \frac{2\pi}{2L^3 \sqrt{k_i^0 k_f^0}} \delta(E_i + k_i^0 - E_f - k_f^0) \\ & \times \sum_{n, s_n} \int d^3x \frac{L da_n dp_z}{2\pi e B 2\pi} \\ & \times \left\{ \frac{[J_{qf, qn}^{(+,+)}(k_f)]^\nu \epsilon_\nu^f [J_{qn, qi}^{(+,+)}(k_i)]^\mu \epsilon_\mu^i}{E_i + k_i^0 - (+1)[E_n^{(1)} - i\Gamma_n/2]} \right\} \\ & + \frac{[J_{qf, qn}^{(+,-)}(k_f)]^\nu \epsilon_\nu^f [J_{qn, qi}^{(-,+)}(k_i)]^\mu \epsilon_\mu^i}{E_i + k_i^0 - (-1)[E_n^{(1)} - i\Gamma_n/2]} \\ & + \frac{[J_{qf, qn}^{(+,+)}(k_i)]^\nu \epsilon_\nu^i [J_{qn, qi}^{(+,+)}(k_f)]^\mu \epsilon_\mu^f}{E_i - k_f^0 - (+1)[E_n^{(2)} - i\Gamma_n/2]} \\ & + \left\{ \frac{[J_{qf, qn}^{(+,-)}(k_i)]^\nu \epsilon_\nu^i [J_{qn, qi}^{(-,+)}(k_f)]^\mu \epsilon_\mu^f}{E_i - k_f^0 - (-1)[E_n^{(2)} - i\Gamma_n/2]} \right\}. \quad (\text{A12}) \end{aligned}$$

The first two terms in the sum correspond to the first diagram (with corresponding intermediate energy $E_n^{(1)}$ given by eq. [A4]), with Γ_n being the decay rate of the intermediate resonant state $n \rightarrow n', s'$ (Graziani 1993).

The expression for the differential cross section:

$$\frac{(2\pi)^3}{L^3} \frac{d\sigma}{d^3k_f} = \int \frac{L^3}{1 - \beta_i \cos \theta_i} \frac{|S_{fi}|^2}{T} \frac{L dp_f}{2\pi} \frac{L da_f}{2\pi e B}.$$

Using the following relations for the averaging over φ :

$$\begin{aligned} \frac{1}{2\pi i^n} \int_0^{2\pi} e^{i(n\varphi + x \cos \varphi)} d\varphi &= J_n(x), \\ \frac{1}{2\pi} \int_0^{2\pi} e^{-i(n-m)\varphi} d\varphi &= \delta_{n,m}, \end{aligned}$$

and the symmetry properties of the J 's:

$$J_{-n}(x) = J_n(-x) = (-1)^n J_n(x),$$

and with the definitions of $Z_{c1}(n)$ and $Z_{c1}(n)$ given in § 4 and the vertex functions, $D_{nf,n}^{(+,+)}(k_f) H_{n,n_i}^{(+,+)}(k_i)$ that follow from Daugherty & Bussard (1980), Sina (1996) and Araya (1996), the explicit form displayed in that section follows.

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